



Assignment Cover Page

Assignment Title:	Mid Term Data Science Project Report		
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1	Md Moynul Islam	22-46984-1	BSc [CSE]	
2			Choose an item.	
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FACULTY COMMENTS	Marks Obtained	
	Total Marks	

Dataset: Introduction

This project focuses on the comprehensive preparation and analysis of the Titanic dataset, which consists of detailed information about passengers aboard the Titanic during its ill-fated voyage. The dataset includes ten columns: Gender, age, sibsp, parch, fare, embarked, class, who, alone, and survived. Each column provides specific and valuable information about the passengers, aiding in various analytical tasks.

The Gender column identifies the gender of the passengers, with values being 'male' or 'female' (character type). The age column records the age of the passengers in integer format. The sibsp column indicates the number of siblings or spouses the passengers had aboard the Titanic, with values ranging from 0 to 4 (integer type). The parch column details the number of parents or children the passengers had on board, also with values ranging from 0 to 4 (integer type). The fare column specifies the fare paid by the passengers for their travel, represented as floating point values (character type).

The embarked column denotes the port of embarkation for the passengers, with three possible values: 'C' for Cherbourg, 'Q' for Queenstown, and 'S' for Southampton (character type). The class column describes the travel class, with values being 'First', 'Second', or 'Third' (character type). The who column categorizes passengers into three categories: 'man', 'woman', and 'child' (character type). The alone column indicates whether a passenger was alone on the ship, with values being TRUE (alone) or FALSE (not alone) (logical type). Finally, the survived column shows whether a passenger survived the disaster, with values being 0 (not survived) or 1 (survived) (integer type).

The primary objective of this project is to prepare the Titanic dataset by addressing missing, noisy, and invalid values. This includes converting numeric attributes into categorical ones and applying normalization techniques. The project begins with visualizations to identify patterns and inform approaches to managing missing data. Methods for correcting invalid data are also explored to enhance dataset quality. Visualization tools like box plots and bar charts will facilitate intelligent analysis. By establishing a strong foundation for data preparation and univariate exploration, this project aims to yield insights into the Titanic disaster, potentially informing future research and decision-making. This report details the methodologies, challenges, and insights gained throughout the project, providing a comprehensive overview of the data preparation and analysis processes.

Dataset: About data

Library Used:

```
library(dplyr)
library(ggplot2)
library(tidyr)
```

Comment: These 3 libraries were used for conducting the necessary operation in this project. 'dplyr' library is used for data manipulation and transformation. 'ggplot2' and 'tidyr' libraires are used for data visualization and manipulation.

Read Data:

```
mydata <- read.csv("D:/8th Semester/INTRODUCTION TO DATA SCIENCE  
[B]/MID/Project/MyProject/Midterm_Project.csv", header = TRUE, sep = ",")
```

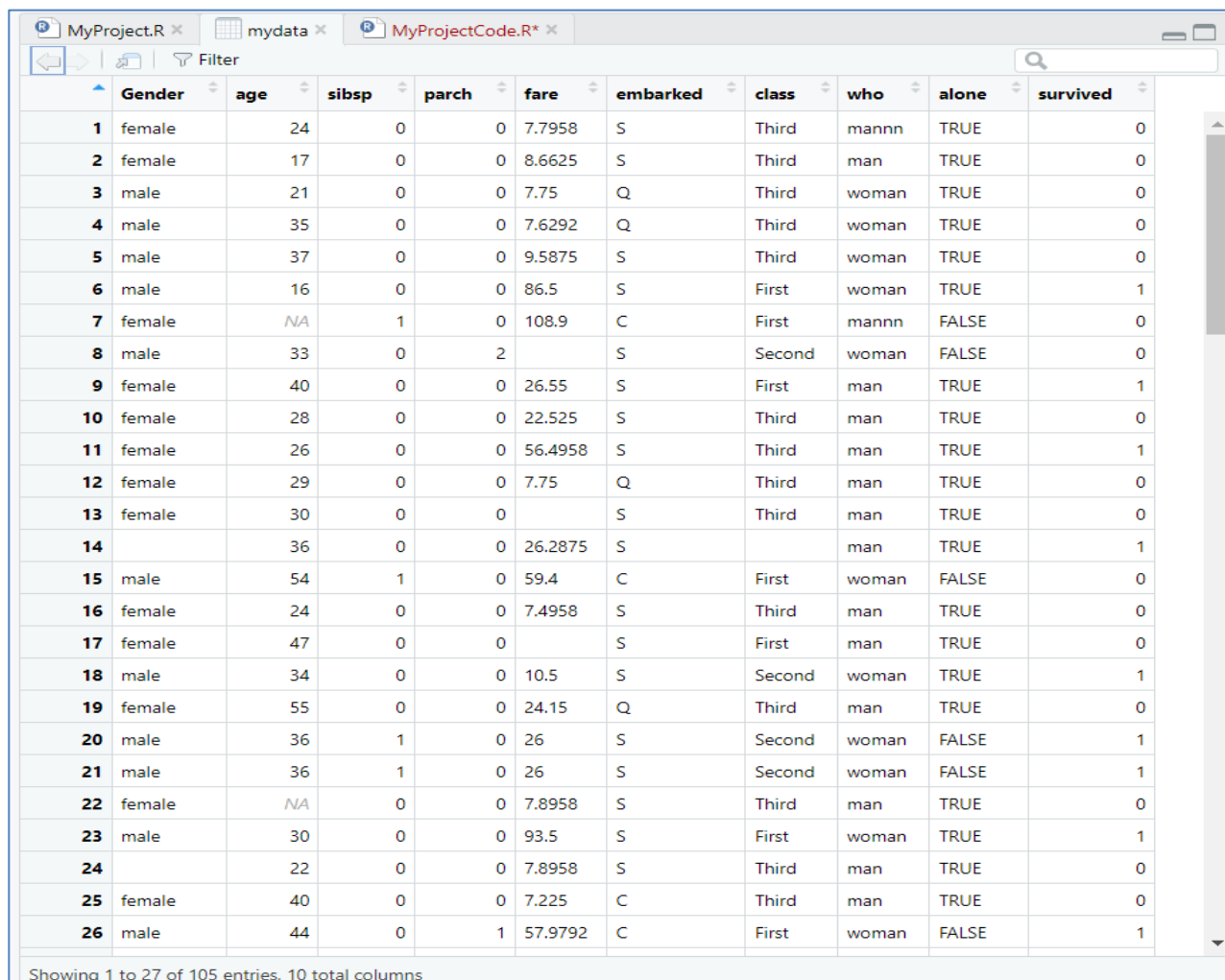
```
View(mydata)
```

```
str(mydata)
```

```
summary(mydata)
```

```
missing_indices <- lapply(mydata, function(x) which(is.na(x)))
```

```
print(missing_indices)
```



	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived
1	female	24	0	0	7.7958	S	Third	mannn	TRUE	0
2	female	17	0	0	8.6625	S	Third	man	TRUE	0
3	male	21	0	0	7.75	Q	Third	woman	TRUE	0
4	male	35	0	0	7.6292	Q	Third	woman	TRUE	0
5	male	37	0	0	9.5875	S	Third	woman	TRUE	0
6	male	16	0	0	86.5	S	First	woman	TRUE	1
7	female	NA	1	0	108.9	C	First	mannn	FALSE	0
8	male	33	0	2		S	Second	woman	FALSE	0
9	female	40	0	0	26.55	S	First	man	TRUE	1
10	female	28	0	0	22.525	S	Third	man	TRUE	0
11	female	26	0	0	56.4958	S	Third	man	TRUE	1
12	female	29	0	0	7.75	Q	Third	man	TRUE	0
13	female	30	0	0		S	Third	man	TRUE	0
14		36	0	0	26.2875	S		man	TRUE	1
15	male	54	1	0	59.4	C	First	woman	FALSE	0
16	female	24	0	0	7.4958	S	Third	man	TRUE	0
17	female	47	0	0		S	First	man	TRUE	0
18	male	34	0	0	10.5	S	Second	woman	TRUE	1
19	female	55	0	0	24.15	Q	Third	man	TRUE	0
20	male	36	1	0	26	S	Second	woman	FALSE	1
21	male	36	1	0	26	S	Second	woman	FALSE	1
22	female	NA	0	0	7.8958	S	Third	man	TRUE	0
23	male	30	0	0	93.5	S	First	woman	TRUE	1
24		22	0	0	7.8958	S	Third	man	TRUE	0
25	female	40	0	0	7.225	C	Third	man	TRUE	0
26	male	44	0	1	57.9792	C	First	woman	FALSE	1

Showing 1 to 27 of 105 entries, 10 total columns

```
> str(mydata)
'data.frame': 105 obs. of 10 variables:
 $ Gender : chr "female" "female" "male" "male" ...
 $ age : num 24 17 21 35 37 16 33 33 40 28 ...
 $ sibsp : int 0 0 0 0 0 0 1 0 0 0 ...
 $ parch : int 0 0 0 0 0 0 0 2 0 0 ...
 $ fare : chr "7.7958" "8.6625" "7.75" "7.6292" ...
 $ embarked: chr "S" "S" "Q" "Q" ...
 $ class : chr "Third" "Third" "Third" "Third" ...
 $ who : chr "man" "man" "woman" "woman" ...
 $ alone : logi TRUE TRUE TRUE TRUE TRUE TRUE ...
 $ survived: int 0 0 0 0 0 1 0 0 1 0 ...
> |
```

```
> summary(mydata)
```

Gender	age	sibsp	parch
Length:105	Min. : 2.00	Min. :0.0000	Min. :0.0000
Class :character	1st Qu.: 23.50	1st Qu.:0.0000	1st Qu.:0.0000
Mode :character	Median : 33.00	Median :0.0000	Median :0.0000
	Mean : 36.25	Mean :0.3495	Mean :0.3398
	3rd Qu.: 40.50	3rd Qu.:1.0000	3rd Qu.:0.0000
	Max. :152.00	Max. :4.0000	Max. :4.0000
	NA's :14	NA's :2	NA's :2

fare	embarked	class	who
Length:105	Length:105	Length:105	Length:105
Class :character	Class :character	Class :character	Class :character
Mode :character	Mode :character	Mode :character	Mode :character

alone	survived
Mode :logical	Min. :0.0000
FALSE:37	1st Qu.:0.0000
TRUE :66	Median :0.0000
NA's :2	Mean :0.3619
	3rd Qu.:1.0000
	Max. :1.0000

```
> missing_indices <- lapply(mydata, function(x) which(is.na(x)))
> print (missing_indices)
```

```
$Gender
integer(0)
```

```
$age
 [1]  7  22  40  61  73  78  83  89  94  98 101 103 104 105
```

```
$sibsp
 [1] 104 105
```

```
$parch
 [1] 104 105
```

```
$fare
integer(0)
```

```
$embarked
integer(0)
```

```
$class
integer(0)
```

```
$who
integer(0)
```

```
$alone
 [1] 104 105
```

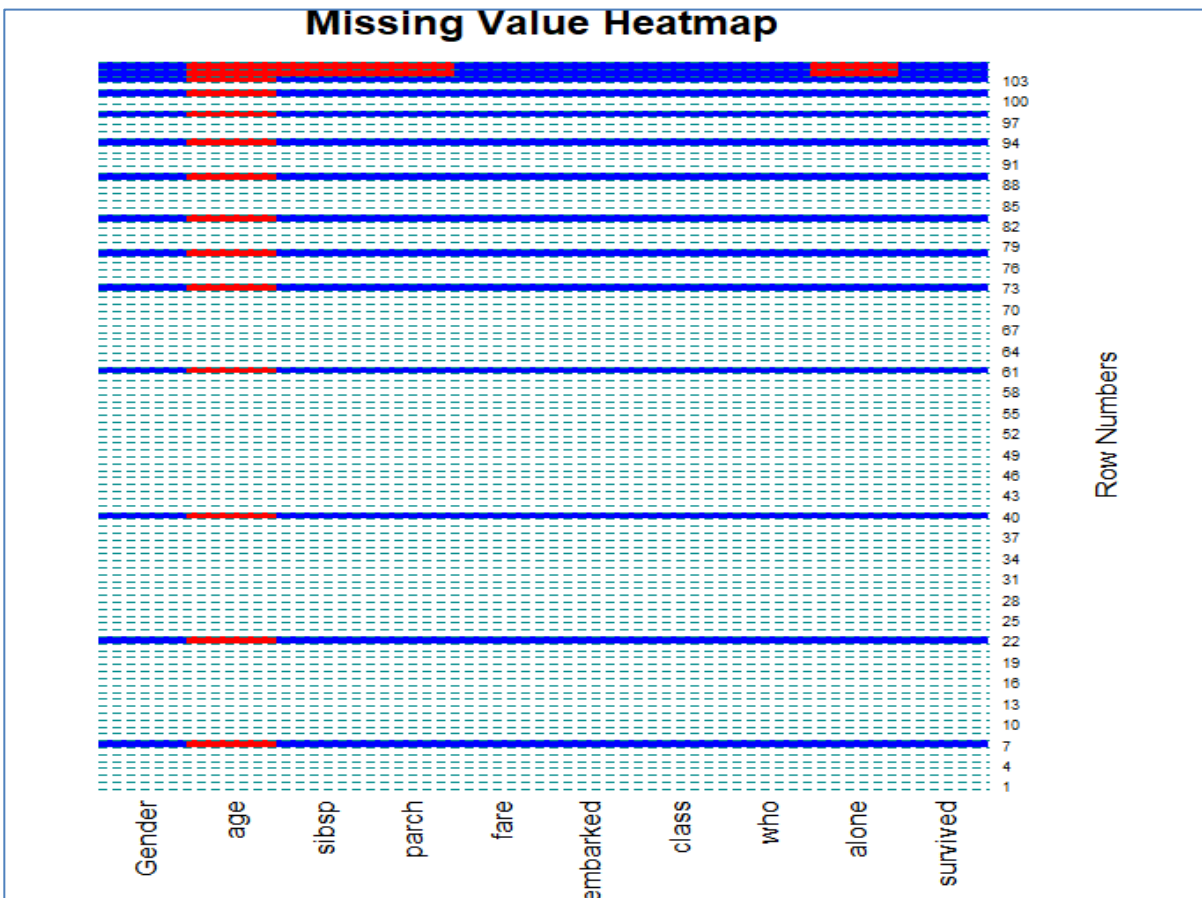
```
$survived
integer(0)
```

```
> |
```

Dataset: Missing Value Heatmap

```
missing_indices <- list(
  Gender = integer(0),
  age = c(7, 22, 40, 61, 73, 78, 83, 89, 94, 98, 101, 103, 104, 105),
  sibsp = c(104,105),
  parch = c(104,105),
  fare = integer(0),
  embarked = integer(0),
  class = integer(0),
  who = integer(0),
  alone = c(104,105),
  survived = integer(0)
)
missing_matrix <- matrix(0, nrow = 105, ncol = length(missing_indices),
  dimnames = list(1:105, names(missing_indices)))
for (col_name in names(missing_indices)) {
  missing_matrix[missing_indices[[col_name]], col_name] <- 1
}

heatmap(missing_matrix, Rowv = NA, Colv = NA, col = c("blue", "red"),
  xlab = "", ylab = "Row Numbers",
  main = "Missing Value Heatmap",
  labRow = 1:105,
  add.expr = { abline(h = seq(0.5, 105.5), col = "cyan4", lty = 2) })
```



Dataset: Data Preparation & Exploration

Column: Gender

- Convert missing Gender to NA

```
mydata$Gender <- ifelse(mydata$Gender == "", NA, mydata$Gender)
View(mydata)
```

Output: Missing Value for Gender column

	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived
1	female	24	0	0	7.7958	S	Third	mannn	TRUE	0
2	female	17	0	0	8.6625	S	Third	man	TRUE	0
3	male	21	0	0	7.75	Q	Third	woman	TRUE	0
4	male	35	0	0	7.6292	Q	Third	woman	TRUE	0
5	male	37	0	0	9.5875	S	Third	woman	TRUE	0
6	male	16	0	0	86.5	S	First	woman	TRUE	1
7	female	NA	1	0	108.9	C	First	mannn	FALSE	0
8	male	33	0	2		S	Second	woman	FALSE	0
9	female	40	0	0	26.55	S	First	man	TRUE	1
10	female	28	0	0	22.525	S	Third	man	TRUE	0
11	female	26	0	0	56.4958	S	Third	man	TRUE	1
12	female	29	0	0	7.75	Q	Third	man	TRUE	0
13	female	30	0	0		S	Third	man	TRUE	0
14	NA	36	0	0	26.2875	S		man	TRUE	1
15	male	54	1	0	59.4	C	First	woman	FALSE	0
16	female	24	0	0	7.4958	S	Third	man	TRUE	0
17	female	47	0	0		S	First	man	TRUE	0
18	male	34	0	0	10.5	S	Second	woman	TRUE	1
19	female	55	0	0	24.15	Q	Third	man	TRUE	0
20	male	36	1	0	26	S	Second	woman	FALSE	1
21	male	36	1	0	26	S	Second	woman	FALSE	1
22	female	NA	0	0	7.8958	S	Third	man	TRUE	0
23	male	30	0	0	93.5	S	First	woman	TRUE	1
24	NA	22	0	0	7.8958	S	Third	man	TRUE	0

- **Missing Value Imputation: Replace with most frequent gender which is Female**

```
people_2_to_152_with_Gender <- mydata %>%filter(age >= 2 & age <= 152, !is.na(Gender))
%>% select(age, Gender)
print(people_2_to_152_with_Gender)
Gender_counts <- table(people_2_to_152_with_Gender$Gender)
most_common_Gender <- names(Gender_counts)[which.max(Gender_counts)]
print(most_common_Gender)
mydata <- mydata %>%
mutate(Gender = ifelse(is.na(Gender), most_common_Gender, Gender))
View(mydata)
```

Output: Most Frequent Common Value for Gender Column

> Gender_counts <- table(people_2_to_152_with_Gender\$Gender)

	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived
1	female	24	0	0	7.7958	S	Third	mannn	TRUE	0
2	female	17	0	0	8.6625	S	Third	man	TRUE	0
3	male	21	0	0	7.75	Q	Third	woman	TRUE	0
4	male	35	0	0	7.6292	Q	Third	woman	TRUE	0
5	male	37	0	0	9.5875	S	Third	woman	TRUE	0
6	male	16	0	0	86.5	S	First	woman	TRUE	1
7	female	NA	1	0	108.9	C	First	mannn	FALSE	0
8	male	33	0	2		S	Second	woman	FALSE	0
9	female	40	0	0	26.55	S	First	man	TRUE	1
10	female	28	0	0	22.525	S	Third	man	TRUE	0
11	female	26	0	0	56.4958	S	Third	man	TRUE	1
12	female	29	0	0	7.75	Q	Third	man	TRUE	0
13	female	30	0	0		S	Third	man	TRUE	0
14	female	36	0	0	26.2875	S		man	TRUE	1
15	male	54	1	0	59.4	C	First	woman	FALSE	0
16	female	24	0	0	7.4958	S	Third	man	TRUE	0
17	female	47	0	0		S	First	man	TRUE	0
18	male	34	0	0	10.5	S	Second	woman	TRUE	1
19	female	55	0	0	24.15	Q	Third	man	TRUE	0
20	male	36	1	0	26	S	Second	woman	FALSE	1
21	male	36	1	0	26	S	Second	woman	FALSE	1
22	female	NA	0	0	7.8958	S	Third	man	TRUE	0
23	male	30	0	0	93.5	S	First	woman	TRUE	1
24	female	22	0	0	7.8958	S	Third	man	TRUE	0
25	female	40	0	0	7.225	C	Third	man	TRUE	0
26	male	44	0	1	57.9792	C	First	woman	FALSE	1

Output: Missing Value replace with most frequent common value "female"

- **Converting Gender to 0 and 1**

```
mydata$Gender <- factor(mydata$Gender,
                        levels = c("male", "female"),
                        labels = c(1, 0))
which(is.na(mydata$Gender))
View (mydata)
```

Output: Gender column value convert into categorical to numerical

	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived
1	0	24	0	0	7.7958	S	Third	mannn	TRUE	0
2	0	17	0	0	8.6625	S	Third	man	TRUE	0
3	1	21	0	0	7.75	Q	Third	woman	TRUE	0
4	1	35	0	0	7.6292	Q	Third	woman	TRUE	0
5	1	37	0	0	9.5875	S	Third	woman	TRUE	0
6	1	16	0	0	86.5	S	First	woman	TRUE	1
7	0	NA	1	0	108.9	C	First	mannn	FALSE	0
8	1	33	0	2		S	Second	woman	FALSE	0
9	0	40	0	0	26.55	S	First	man	TRUE	1
10	0	28	0	0	22.525	S	Third	man	TRUE	0
11	0	26	0	0	56.4958	S	Third	man	TRUE	1
12	0	29	0	0	7.75	Q	Third	man	TRUE	0
13	0	30	0	0		S	Third	man	TRUE	0
14	0	36	0	0	26.2875	S		man	TRUE	1
15	1	54	1	0	59.4	C	First	woman	FALSE	0
16	0	24	0	0	7.4958	S	Third	man	TRUE	0
17	0	47	0	0		S	First	man	TRUE	0
18	1	34	0	0	10.5	S	Second	woman	TRUE	1
19	0	55	0	0	24.15	Q	Third	man	TRUE	0
20	1	36	1	0	26	S	Second	woman	FALSE	1
21	1	36	1	0	26	S	Second	woman	FALSE	1
22	0	NA	0	0	7.8958	S	Third	man	TRUE	0
23	1	30	0	0	93.5	S	First	woman	TRUE	1
24	0	22	0	0	7.8958	S	Third	man	TRUE	0
25	0	40	0	0	7.225	C	Third	man	TRUE	0
26	1	44	0	1	57.9792	C	First	woman	FALSE	1

Comment: Missing values were retrieved using Most Frequent Value method which is 'female'. The values of 'Gender' column were categorical initially. Later, it was converted to numerical values as '0' and '1', male being '1' and female being '0'.

Column: Age

- **Data Transformation: Rounding Age Values**

```
mydata$age <- round(mydata$age)
```

- **Missing Value Imputation: Converting Age to Integer and Replacing NA Values with Median**

```
age_median <- round(median(mydata$age, na.rm = TRUE))  
mydata$age[is.na(mydata$age)] <- age_median
```

- **Outlier Detection and Removal with Interquartile Range (IQR) Method**

```
Q1 <- quantile(mydata$age[mydata$age >= 2 & mydata$age <= 70], 0.25)  
Q3 <- quantile(mydata$age[mydata$age >= 2 & mydata$age <= 70], 0.75)  
IQR_value <- Q3 - Q1  
  
threshold <- 1.5  
  
outlier_condition <- (mydata$age < (Q1 - threshold * IQR_value)) | (mydata$age > (Q3 +  
threshold * IQR_value))  
  
mydata <- mydata[!outlier_condition, ]  
View(mydata)
```

- **Serial Update After Removing Outliers**

```
mydata <- mydata %>%  
  filter(!is.na(age)) %>%  
  mutate(row_number = row_number())
```

Output: Removing outliers and missing value imputation with median value.

	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived
1	0	24	0	0	7.7958	S	Third	mannn	TRUE	0
2	0	17	0	0	8.6625	S	Third	man	TRUE	0
3	1	21	0	0	7.75	Q	Third	woman	TRUE	0
4	1	35	0	0	7.6292	Q	Third	woman	TRUE	0
5	1	37	0	0	9.5875	S	Third	woman	TRUE	0
6	1	16	0	0	86.5	S	First	woman	TRUE	1
7	0	33	1	0	108.9	C	First	mannn	FALSE	0
8	1	33	0	2		S	Second	woman	FALSE	0
9	0	40	0	0	26.55	S	First	man	TRUE	1
10	0	28	0	0	22.525	S	Third	man	TRUE	0
11	0	26	0	0	56.4958	S	Third	man	TRUE	1
12	0	29	0	0	7.75	Q	Third	man	TRUE	0
13	0	30	0	0		S	Third	man	TRUE	0
14	0	36	0	0	26.2875	S		man	TRUE	1
15	1	54	1	0	59.4	C	First	woman	FALSE	0
16	0	24	0	0	7.4958	S	Third	man	TRUE	0
17	0	47	0	0		S	First	man	TRUE	0
18	1	34	0	0	10.5	S	Second	woman	TRUE	1
19	0	55	0	0	24.15	Q	Third	man	TRUE	0
20	1	36	1	0	26	S	Second	woman	FALSE	1
21	1	36	1	0	26	S	Second	woman	FALSE	1
22	0	33	0	0	7.8958	S	Third	man	TRUE	0
23	1	30	0	0	93.5	S	First	woman	TRUE	1
24	0	22	0	0	7.8958	S	Third	man	TRUE	0
25	0	40	0	0	7.225	C	Third	man	TRUE	0
26	1	44	0	1	57.9792	C	First	woman	FALSE	1
27	0	38	0	0	7.2292	C	Third	man	TRUE	0

Comment: First, the fractional values of 'Age' column were rounded, and the value type were converted to Numeric. Later, Outliers were detected using Interquartile Range (IQR) method. As the serial number of the columns became imbalanced, it was updated as a numeric serial number.

Column: sibsp

▪ Changing Missing Values with Most Frequent Value

```
people_2_to_152_with_sibsp <- mydata %>%filter(age >= 2 & age <= 152, !is.na(sibsp)) %>%
select(age, sibsp)
print(people_2_to_152_with_sibsp)
sibsp_counts <- table(people_2_to_152_with_sibsp$sibsp)
most_common_sibsp <- names(sibsp_counts)[which.max(sibsp_counts)]
print(most_common_sibsp)
mydata <- mydata %>%
  mutate(sibsp = ifelse(is.na(sibsp), most_common_sibsp, sibsp))
View(mydata)
```

Output: Most Frequent Common Value for sibsp Column

```
> print(most_common_sibsp)
[1] "0"
> |
```

Output: Missing Value replace with most frequent common value “0”

	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived	row_number
1	0	24	0	0	7.7958	S	Third	mann	TRUE	0	1
2	0	17	0	0	8.6625	S	Third	man	TRUE	0	2
3	1	21	0	0	7.75	Q	Third	woman	TRUE	0	3
4	1	35	0	0	7.6292	Q	Third	woman	TRUE	0	4
5	1	37	0	0	9.5875	S	Third	woman	TRUE	0	5
6	1	16	0	0	86.5	S	First	woman	TRUE	1	6
7	0	33	1	0	108.9	C	First	mann	FALSE	0	7
8	1	33	0	2		S	Second	woman	FALSE	0	8
9	0	40	0	0	26.55	S	First	man	TRUE	1	9
10	0	28	0	0	22.525	S	Third	man	TRUE	0	10
11	0	26	0	0	56.4958	S	Third	man	TRUE	1	11
12	0	29	0	0	7.75	Q	Third	man	TRUE	0	12
13	0	30	0	0		S	Third	man	TRUE	0	13
14	0	36	0	0	26.2875	S		man	TRUE	1	14
15	1	54	1	0	59.4	C	First	woman	FALSE	0	15
16	0	24	0	0	7.4958	S	Third	man	TRUE	0	16
17	0	47	0	0		S	First	man	TRUE	0	17
18	1	34	0	0	10.5	S	Second	woman	TRUE	1	18
19	0	55	0	0	24.15	Q	Third	man	TRUE	0	19
20	1	36	1	0	26	S	Second	woman	FALSE	1	20
21	1	36	1	0	26	S	Second	woman	FALSE	1	21
22	0	33	0	0	7.8958	S	Third	man	TRUE	0	22
23	1	30	0	0	93.5	S	First	woman	TRUE	1	23
24	0	22	0	0	7.8958	S	Third	man	TRUE	0	24
25	0	40	0	0	7.225	C	Third	man	TRUE	0	25
26	1	44	0	1	57.9792	C	First	woman	FALSE	1	26

Comment: Most frequent value was print and missing values were replaced with it.

Column: parch

▪ Changing Missing Value

```
people_2_to_152_with_parch <- mydata %>%filter(age >= 2 & age <= 152, !is.na(parch)) %>%
select(age, parch)
print(people_2_to_152_with_parch)
parch_counts <- table(people_2_to_152_with_parch$parch)
most_common_parch <- names(parch_counts)[which.max(parch_counts)]
print(most_common_parch)
mydata <- mydata %>%
  mutate(parch = ifelse(is.na(parch), most_common_parch, parch))
View(mydata)
```

Output: Most Frequent Common Value for parch Column

```
> print(most_common_parch)
[1] "0"
> |
```

Output: Missing Value replace with most frequent common value “0”

	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived	row_number
1	0	24	0	0	7.7958	S	Third	mannn	TRUE	0	1
2	0	17	0	0	8.6625	S	Third	man	TRUE	0	2
3	1	21	0	0	7.75	Q	Third	woman	TRUE	0	3
4	1	35	0	0	7.6292	Q	Third	woman	TRUE	0	4
5	1	37	0	0	9.5875	S	Third	woman	TRUE	0	5
6	1	16	0	0	86.5	S	First	woman	TRUE	1	6
7	0	33	1	0	108.9	C	First	mannn	FALSE	0	7
8	1	33	0	2		S	Second	woman	FALSE	0	8
9	0	40	0	0	26.55	S	First	man	TRUE	1	9
10	0	28	0	0	22.525	S	Third	man	TRUE	0	10
11	0	26	0	0	56.4958	S	Third	man	TRUE	1	11
12	0	29	0	0	7.75	Q	Third	man	TRUE	0	12
13	0	30	0	0		S	Third	man	TRUE	0	13
14	0	36	0	0	26.2875	S		man	TRUE	1	14
15	1	54	1	0	59.4	C	First	woman	FALSE	0	15
16	0	24	0	0	7.4958	S	Third	man	TRUE	0	16
17	0	47	0	0		S	First	man	TRUE	0	17
18	1	34	0	0	10.5	S	Second	woman	TRUE	1	18
19	0	55	0	0	24.15	Q	Third	man	TRUE	0	19
20	1	36	1	0	26	S	Second	woman	FALSE	1	20
21	1	36	1	0	26	S	Second	woman	FALSE	1	21
22	0	33	0	0	7.8958	S	Third	man	TRUE	0	22
23	1	30	0	0	93.5	S	First	woman	TRUE	1	23
24	0	22	0	0	7.8958	S	Third	man	TRUE	0	24
25	0	40	0	0	7.225	C	Third	man	TRUE	0	25
26	1	44	0	1	57.9792	C	First	woman	FALSE	1	26
27	0	28	0	0	7.2292	C	Third	man	TRUE	0	27
28	0	41	0	0	7.75	Q	Third	man	TRUE	0	28

Comment: Most frequent value was print and missing values were replaced with it.

Column: Fare

▪ Converting Missing Values to NA

```
mydata$fare <- ifelse(mydata$fare == "", NA, mydata$fare)
View(mydata)
```

▪ Most Frequent Value

```
people_2_to_152_with_fare <- mydata %>%filter(age >= 2 & age <= 152, !is.na(fare)) %>%
select(age, fare)
print(people_2_to_152_with_fare)
fare_counts <- table(people_2_to_152_with_fare$fare)
most_common_fare <- names(fare_counts)[which.max(fare_counts)]
print(most_common_fare)
mydata <- mydata %>%
  mutate(fare = ifelse(is.na(fare), most_common_fare, fare))
View(mydata)
```

▪ Converting Data Type to Numeric

```
mydata$fare <- as.numeric(as.character(mydata$fare))
print(mydata$fare)
mydata$fare <- round(mydata$fare)
View(mydata)
```

Output

	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived	row_number
1	0	24	0	0	8	S	Third	mannn	TRUE	0	1
2	0	17	0	0	9	S	Third	man	TRUE	0	2
3	1	21	0	0	8	Q	Third	woman	TRUE	0	3
4	1	35	0	0	8	Q	Third	woman	TRUE	0	4
5	1	37	0	0	10	S	Third	woman	TRUE	0	5
6	1	16	0	0	86	S	First	woman	TRUE	1	6
7	0	33	1	0	109	C	First	mannn	FALSE	0	7
8	1	33	0	2	NA	S	Second	woman	FALSE	0	8
9	0	40	0	0	27	S	First	man	TRUE	1	9
10	0	28	0	0	23	S	Third	man	TRUE	0	10
11	0	26	0	0	56	S	Third	man	TRUE	1	11
12	0	29	0	0	8	Q	Third	man	TRUE	0	12
13	0	30	0	0	NA	S	Third	man	TRUE	0	13
14	0	36	0	0	26	S		man	TRUE	1	14
15	1	54	1	0	59	C	First	woman	FALSE	0	15
16	0	24	0	0	7	S	Third	man	TRUE	0	16
17	0	47	0	0	NA	S	First	man	TRUE	0	17
18	1	34	0	0	10	S	Second	woman	TRUE	1	18
19	0	55	0	0	24	Q	Third	man	TRUE	0	19
20	1	36	1	0	26	S	Second	woman	FALSE	1	20
21	1	36	1	0	26	S	Second	woman	FALSE	1	21
22	0	33	0	0	8	S	Third	man	TRUE	0	22
23	1	30	0	0	94	S	First	woman	TRUE	1	23
24	0	22	0	0	8	S	Third	man	TRUE	0	24
25	0	40	0	0	7	C	Third	man	TRUE	0	25
26	1	44	0	1	58	C	First	woman	FALSE	1	26
27	0	28	0	0	7	C	Third	man	TRUE	0	27
28	0	41	0	0	8	Q	Third	man	TRUE	0	28

▪ Normalization in Fare

```
fare_column <- mydata$fare
normalized_fare <- (fare_column - min(fare_column)) / (max(fare_column) - min(fare_column))
mydata$fare <- normalized_fare
head(mydata)
print(mydata)
```

Output:

	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived	row_number
1	0	24	0	0	0.004524887	S	Third	mannn	TRUE	0	1
2	0	17	0	0	0.009049774	S	Third	man	TRUE	0	2
3	1	21	0	0	0.004524887	Q	Third	woman	TRUE	0	3
4	1	35	0	0	0.004524887	Q	Third	woman	TRUE	0	4
5	1	37	0	0	0.013574661	S	Third	woman	TRUE	0	5
6	1	16	0	0	0.357466063	S	First	woman	TRUE	1	6
7	0	33	1	0	0.461538462	C	First	mannn	FALSE	0	7
8	1	33	0	2	0.004524887	S	Second	woman	FALSE	0	8
9	0	40	0	0	0.090497738	S	First	man	TRUE	1	9
10	0	28	0	0	0.072398190	S	Third	man	TRUE	0	10
11	0	26	0	0	0.221719457	S	Third	man	TRUE	1	11
12	0	29	0	0	0.004524887	Q	Third	man	TRUE	0	12
13	0	30	0	0	0.004524887	S	Third	man	TRUE	0	13
14	0	36	0	0	0.085972851	S		man	TRUE	1	14
15	1	54	1	0	0.235294118	C	First	woman	FALSE	0	15
16	0	24	0	0	0.000000000	S	Third	man	TRUE	0	16
17	0	47	0	0	0.004524887	S	First	man	TRUE	0	17
18	1	34	0	0	0.013574661	S	Second	woman	TRUE	1	18
19	0	55	0	0	0.076923077	Q	Third	man	TRUE	0	19
20	1	36	1	0	0.085972851	S	Second	woman	FALSE	1	20
21	1	36	1	0	0.085972851	S	Second	woman	FALSE	1	21
22	0	33	0	0	0.004524887	S	Third	man	TRUE	0	22
23	1	30	0	0	0.393665158	S	First	woman	TRUE	1	23
24	0	22	0	0	0.004524887	S	Third	man	TRUE	0	24
25	0	40	0	0	0.000000000	C	Third	man	TRUE	0	25
26	1	44	0	1	0.230769231	C	First	woman	FALSE	1	26

Comment: First, missing values were converted to NA and then it was replaced with most frequent value. Then, data type was converted to Numeric. Normalization was applied in 'fare' column at last as the values of this column were significantly larger compared to the rest of the columns.

Column: Embarked

- **Convert to NA**

```
mydata$embarked <- ifelse(mydata$embarked == "", NA, mydata$embarked)
View(mydata)
```

- **Most Frequent Value**

```
people_2_to_152_with_embarked <- mydata %>%filter(age >= 2 & age <= 152,
!is.na(embarked)) %>% select(age, embarked)
print(people_2_to_152_with_embarked)
embarked_counts <- table(people_2_to_152_with_embarked$embarked)
most_common_embarked <- names(embarked_counts)[which.max(embarked_counts)]
print(most_common_embarked)
mydata <- mydata %>%
  mutate(embarked = ifelse(is.na(embarked), most_common_embarked, embarked))
View(mydata)
```

- **Converting Embarked to S=1, C=2, Q=3**

```
mydata$embarked <- factor(mydata$embarked,
  levels = c("S", "C", "Q"),
  labels = c(1, 2, 3))
which(is.na(mydata$embarked))
View (mydata)
```

Output: Converting Embarked to S=1, C=2, Q=3

	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived	row_number
1	0	24	0	0	7.7958	1	Third	mannn	TRUE	0	1
2	0	17	0	0	8.6625	1	Third	man	TRUE	0	2
3	1	21	0	0	7.75	3	Third	woman	TRUE	0	3
4	1	35	0	0	7.6292	3	Third	woman	TRUE	0	4
5	1	37	0	0	9.5875	1	Third	woman	TRUE	0	5
6	1	16	0	0	86.5	1	First	woman	TRUE	1	6
7	0	33	1	0	108.9	2	First	mannn	FALSE	0	7
8	1	33	0	2	26	1	Second	woman	FALSE	0	8
9	0	40	0	0	26.55	1	First	man	TRUE	1	9
10	0	28	0	0	22.525	1	Third	man	TRUE	0	10
11	0	26	0	0	56.4958	1	Third	man	TRUE	1	11
12	0	29	0	0	7.75	3	Third	man	TRUE	0	12
13	0	30	0	0	26	1	Third	man	TRUE	0	13
14	0	36	0	0	26.2875	1		man	TRUE	1	14
15	1	54	1	0	59.4	2	First	woman	FALSE	0	15
16	0	24	0	0	7.4958	1	Third	man	TRUE	0	16
17	0	47	0	0	26	1	First	man	TRUE	0	17
18	1	34	0	0	10.5	1	Second	woman	TRUE	1	18
19	0	55	0	0	24.15	3	Third	man	TRUE	0	19
20	1	36	1	0	26	1	Second	woman	FALSE	1	20
21	1	36	1	0	26	1	Second	woman	FALSE	1	21
22	0	33	0	0	7.8958	1	Third	man	TRUE	0	22
23	1	30	0	0	93.5	1	First	woman	TRUE	1	23
24	0	22	0	0	7.8958	1	Third	man	TRUE	0	24
25	0	40	0	0	7.225	2	Third	man	TRUE	0	25
26	1	44	0	1	57.9792	2	First	woman	FALSE	1	26
27	0	28	0	0	7.2292	2	Third	man	TRUE	0	27
28	0	41	0	0	7.75	3	Third	man	TRUE	0	28

Comment: First, the data type was converted to Numeric. The, the missing values were replaced with most frequent value. The 3 categorical values in the 'Embarked' column were converted into Numerical as S=1, C=2, Q=3.

Column: Class

▪ Convert to NA

```
mydata$class <- ifelse(mydata$class == "", NA, mydata$class)
View(mydata)
```

▪ Most Frequent Value

```
people_2_to_152_with_class <- mydata %>% filter(age >= 2 & age <= 152, !is.na(class))
%>% select(age, class)
print(people_2_to_152_with_class)
class_counts <- table(people_2_to_152_with_class$class)
most_common_class <- names(class_counts)[which.max(class_counts)]
print(most_common_class)
mydata <- mydata %>%
  mutate(class = ifelse(is.na(class), most_common_class, class))
View(mydata)
```

- **Converting Class to First=1, Second=2, Third=3**

```
mydata$class <- factor(mydata$class,
                        levels = c("First", "Second", "Third"),
                        labels = c(1, 2, 3))
which(is.na(mydata$class))
View (mydata)
```

Output:

	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived	row_number
1	0	24	0	0	7.7958	1	3	mannn	TRUE	0	1
2	0	17	0	0	8.6625	1	3	man	TRUE	0	2
3	1	21	0	0	7.75	3	3	woman	TRUE	0	3
4	1	35	0	0	7.6292	3	3	woman	TRUE	0	4
5	1	37	0	0	9.5875	1	3	woman	TRUE	0	5
6	1	16	0	0	86.5	1	1	woman	TRUE	1	6
7	0	33	1	0	108.9	2	1	mannn	FALSE	0	7
8	1	33	0	2	26	1	2	woman	FALSE	0	8
9	0	40	0	0	26.55	1	1	man	TRUE	1	9
10	0	28	0	0	22.525	1	3	man	TRUE	0	10
11	0	26	0	0	56.4958	1	3	man	TRUE	1	11
12	0	29	0	0	7.75	3	3	man	TRUE	0	12
13	0	30	0	0	26	1	3	man	TRUE	0	13
14	0	36	0	0	26.2875	1	3	man	TRUE	1	14
15	1	54	1	0	59.4	2	1	woman	FALSE	0	15
16	0	24	0	0	7.4958	1	3	man	TRUE	0	16
17	0	47	0	0	26	1	1	man	TRUE	0	17
18	1	34	0	0	10.5	1	2	woman	TRUE	1	18
19	0	55	0	0	24.15	3	3	man	TRUE	0	19
20	1	36	1	0	26	1	2	woman	FALSE	1	20
21	1	36	1	0	26	1	2	woman	FALSE	1	21
22	0	33	0	0	7.8958	1	3	man	TRUE	0	22
23	1	30	0	0	93.5	1	1	woman	TRUE	1	23
24	0	22	0	0	7.8958	1	3	man	TRUE	0	24
25	0	40	0	0	7.225	2	3	man	TRUE	0	25
26	1	44	0	1	57.9792	2	1	woman	FALSE	1	26
27	0	28	0	0	7.2292	2	3	man	TRUE	0	27
28	0	41	0	0	7.75	3	3	man	TRUE	0	28

Comment: First, missing values were converted to NA and then it was replaced with most frequent value. The 3 categorical values in the 'class' column were converted into Numerical as First=1, Second=2, Third=3.

Column: Who

- **Unique value of who column**

```
unique_values <- unique(mydata$who)
print(unique_values)
mydata$who <- gsub("mannn", "man", mydata$who)
```

- **Convert to NA**

```
mydata$who <- ifelse(mydata$who == "", NA, mydata$who)
View(mydata)
```

- **Most Frequent Value**

```
people_2_to_152_with_who <- mydata %>%filter(age >= 2 & age <= 152, !is.na(who)) %>%
select(age, who)
print(people_2_to_152_with_who)
who_counts <- table(people_2_to_152_with_who$who)
most_common_who <- names(who_counts)[which.max(who_counts)]
print(most_common_who)
mydata <- mydata %>%
  mutate(who = ifelse(is.na(who), most_common_who, who))
View(mydata)
```

- **Converting Who to 1, 2 and 3**

```
mydata$who <- factor(mydata$who,
  levels = c("man", "woman", "child"),
  labels = c(1, 2, 3))
which(is.na(mydata$who))
View (mydata)
```

Output: Convert who column to man=1, woman=2, child=3

	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived	row_number
1	0	24	0	0	7.7958	1	3	1	TRUE	0	1
2	0	17	0	0	8.6625	1	3	1	TRUE	0	2
3	1	21	0	0	7.75	3	3	2	TRUE	0	3
4	1	35	0	0	7.6292	3	3	2	TRUE	0	4
5	1	37	0	0	9.5875	1	3	2	TRUE	0	5
6	1	16	0	0	86.5	1	1	2	TRUE	1	6
7	0	33	1	0	108.9	2	1	1	FALSE	0	7
8	1	33	0	2	26	1	2	2	FALSE	0	8
9	0	40	0	0	26.55	1	1	1	TRUE	1	9
10	0	28	0	0	22.525	1	3	1	TRUE	0	10
11	0	26	0	0	56.4958	1	3	1	TRUE	1	11
12	0	29	0	0	7.75	3	3	1	TRUE	0	12
13	0	30	0	0	26	1	3	1	TRUE	0	13
14	0	36	0	0	26.2875	1	3	1	TRUE	1	14
15	1	54	1	0	59.4	2	1	2	FALSE	0	15
16	0	24	0	0	7.4958	1	3	1	TRUE	0	16
17	0	47	0	0	26	1	1	1	TRUE	0	17
18	1	34	0	0	10.5	1	2	2	TRUE	1	18
19	0	55	0	0	24.15	3	3	1	TRUE	0	19
20	1	36	1	0	26	1	2	2	FALSE	1	20
21	1	36	1	0	26	1	2	2	FALSE	1	21
22	0	33	0	0	7.8958	1	3	1	TRUE	0	22
23	1	30	0	0	93.5	1	1	2	TRUE	1	23
24	0	22	0	0	7.8958	1	3	1	TRUE	0	24
25	0	40	0	0	7.225	2	3	1	TRUE	0	25
26	1	44	0	1	57.9792	2	1	2	FALSE	1	26
27	0	28	0	0	7.2292	2	3	1	TRUE	0	27
28	0	41	0	0	7.75	3	3	1	TRUE	0	28

Comment: First, the unique values of 'who' column were identified and invalid values like mannn were converted to man. Then, missing values were converted to NA and then it was replaced with most frequent value. Finally, the 3 categorical values in the 'who' column were converted into Numerical as man=1, woman=2, child=3.

Column: alone

▪ Missing Value

```
mydata$alone <- ifelse(mydata$alone == "", NA, mydata$alone)
View(mydata)
```

▪ Most Frequent Value

```
people_2_to_152_with_alone <- mydata %>%filter(age >= 2 & age <= 152, !is.na(alone)) %>%
select(age, alone)
print(people_2_to_152_with_who)
alone_counts <- table(people_2_to_152_with_alone$alone)
most_common_alone <- names(alone_counts)[which.max(alone_counts)]
print(most_common_alone)
mydata <- mydata %>%
mutate(alone = ifelse(is.na(alone), most_common_alone, alone))
View(mydata)
```

- **Converting Alone to 0 and 1**

```
mydata$alone <- factor(mydata$alone,
                        levels = c("TRUE", "FALSE"),
                        labels = c(1, 0))
which(is.na(mydata$alone))
View (mydata)
```

Output: Convert alone column to TRUE=1, FALSE=0

	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived	row_number
1	0	24	0	0	7.7958	1	3	1	1	0	1
2	0	17	0	0	8.6625	1	3	1	1	0	2
3	1	21	0	0	7.75	3	3	2	1	0	3
4	1	35	0	0	7.6292	3	3	2	1	0	4
5	1	37	0	0	9.5875	1	3	2	1	0	5
6	1	16	0	0	86.5	1	1	2	1	1	6
7	0	33	1	0	108.9	2	1	1	0	0	7
8	1	33	0	2	26	1	2	2	0	0	8
9	0	40	0	0	26.55	1	1	1	1	1	9
10	0	28	0	0	22.525	1	3	1	1	0	10
11	0	26	0	0	56.4958	1	3	1	1	1	11
12	0	29	0	0	7.75	3	3	1	1	0	12
13	0	30	0	0	26	1	3	1	1	0	13
14	0	36	0	0	26.2875	1	3	1	1	1	14
15	1	54	1	0	59.4	2	1	2	0	0	15
16	0	24	0	0	7.4958	1	3	1	1	0	16
17	0	47	0	0	26	1	1	1	1	0	17
18	1	34	0	0	10.5	1	2	2	1	1	18
19	0	55	0	0	24.15	3	3	1	1	0	19
20	1	36	1	0	26	1	2	2	0	1	20
21	1	36	1	0	26	1	2	2	0	1	21
22	0	33	0	0	7.8958	1	3	1	1	0	22
23	1	30	0	0	93.5	1	1	2	1	1	23
24	0	22	0	0	7.8958	1	3	1	1	0	24
25	0	40	0	0	7.225	2	3	1	1	0	25
26	1	44	0	1	57.9792	2	1	2	0	1	26
27	0	28	0	0	7.2292	2	3	1	1	0	27
28	0	41	0	0	7.75	3	3	1	1	0	28

Comment: First, missing values were converted to NA and then it was replaced with most frequent value. The 2 logical values in the 'alone' column was converted into Numerical as TRUE=1, FALSE=0.

Column: survived

▪ Changing Missing Value for Survived column

```
anyNA(mydata$survived)
```

Output: Checking if Survived Column has any missing data

```
> #Checking if Survived Column has any missing data
> anyNA(mydata$survived)
[1] FALSE
>
```

Output: Survived column doesn't have any missing data

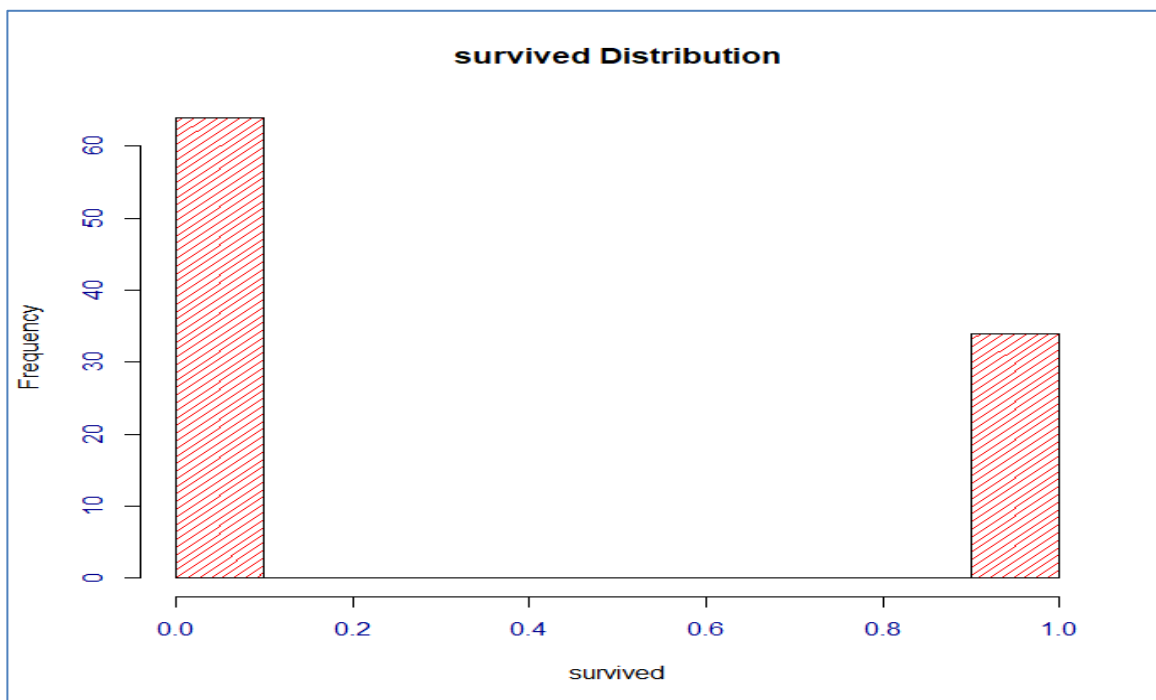
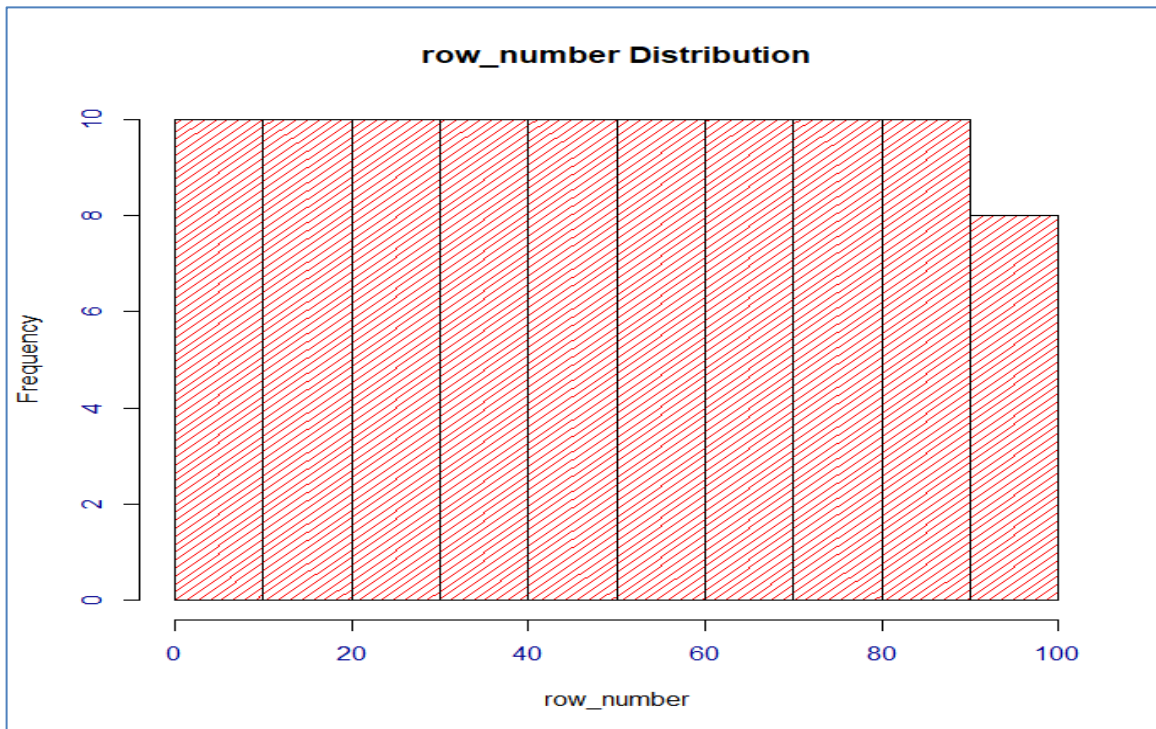
	Gender	age	sibsp	parch	fare	embarked	class	who	alone	survived
1	0	24	0	0	0.004524887	1	3	1	1	0
2	0	17	0	0	0.009049774	1	3	1	1	0
3	1	21	0	0	0.004524887	3	3	2	1	0
4	1	35	0	0	0.004524887	3	3	2	1	0
5	1	37	0	0	0.013574661	1	3	2	1	0
6	1	16	0	0	0.357466063	1	1	2	1	1
7	0	33	1	0	0.461538462	2	1	1	0	0
8	1	33	0	2	0.004524887	1	2	2	0	0
9	0	40	0	0	0.090497738	1	1	1	1	1
10	0	28	0	0	0.072398190	1	3	1	1	0
11	0	26	0	0	0.221719457	1	3	1	1	1
12	0	29	0	0	0.004524887	3	3	1	1	0
13	0	30	0	0	0.004524887	1	3	1	1	0
14	0	36	0	0	0.085972851	1	3	1	1	1
15	1	54	1	0	0.235294118	2	1	2	0	0
16	0	24	0	0	0.000000000	1	3	1	1	0
17	0	47	0	0	0.004524887	1	1	1	1	0
18	1	34	0	0	0.013574661	1	2	2	1	1
19	0	55	0	0	0.076923077	3	3	1	1	0
20	1	36	1	0	0.085972851	1	2	2	0	1
21	1	36	1	0	0.085972851	1	2	2	0	1
22	0	33	0	0	0.004524887	1	3	1	1	0
23	1	30	0	0	0.393665158	1	1	2	1	1
24	0	22	0	0	0.004524887	1	3	1	1	0
25	0	40	0	0	0.000000000	2	3	1	1	0
26	1	44	0	1	0.230769231	2	1	2	0	1
27	0	28	0	0	0.000000000	2	3	1	1	0
28	0	41	0	0	0.004524887	3	3	1	1	0

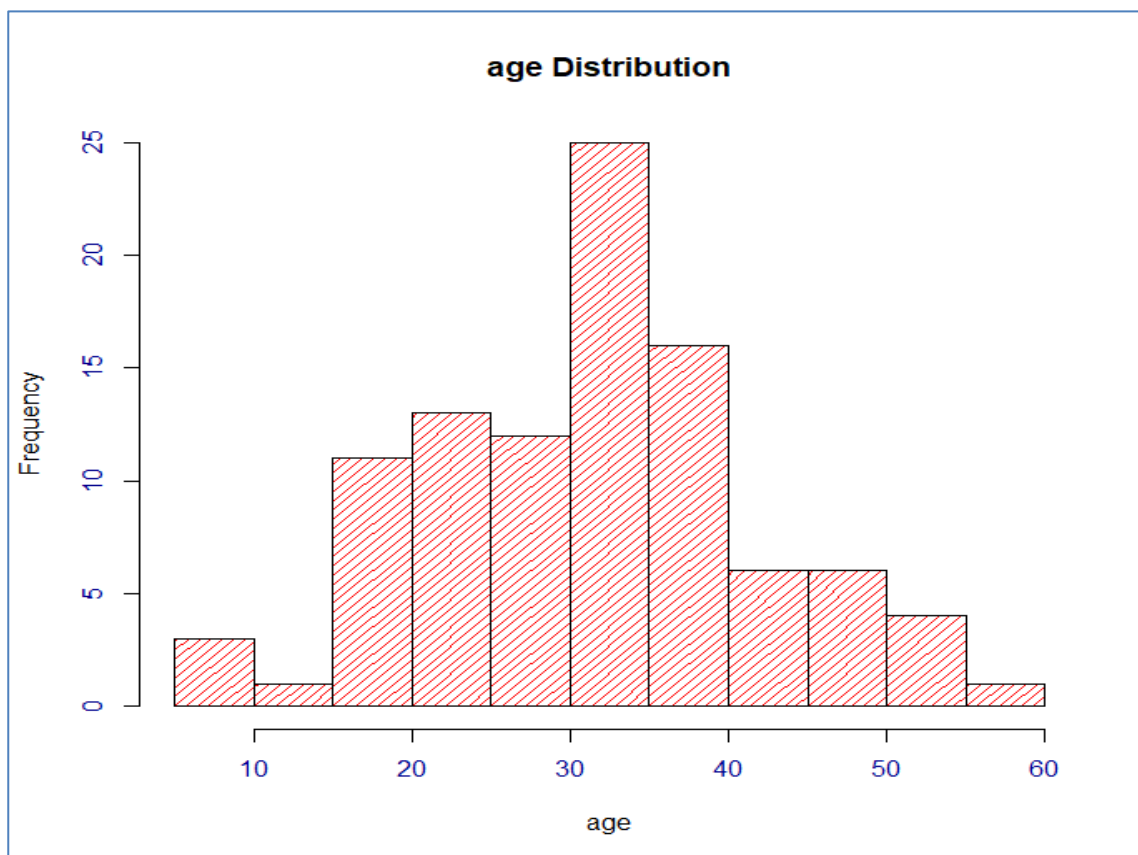
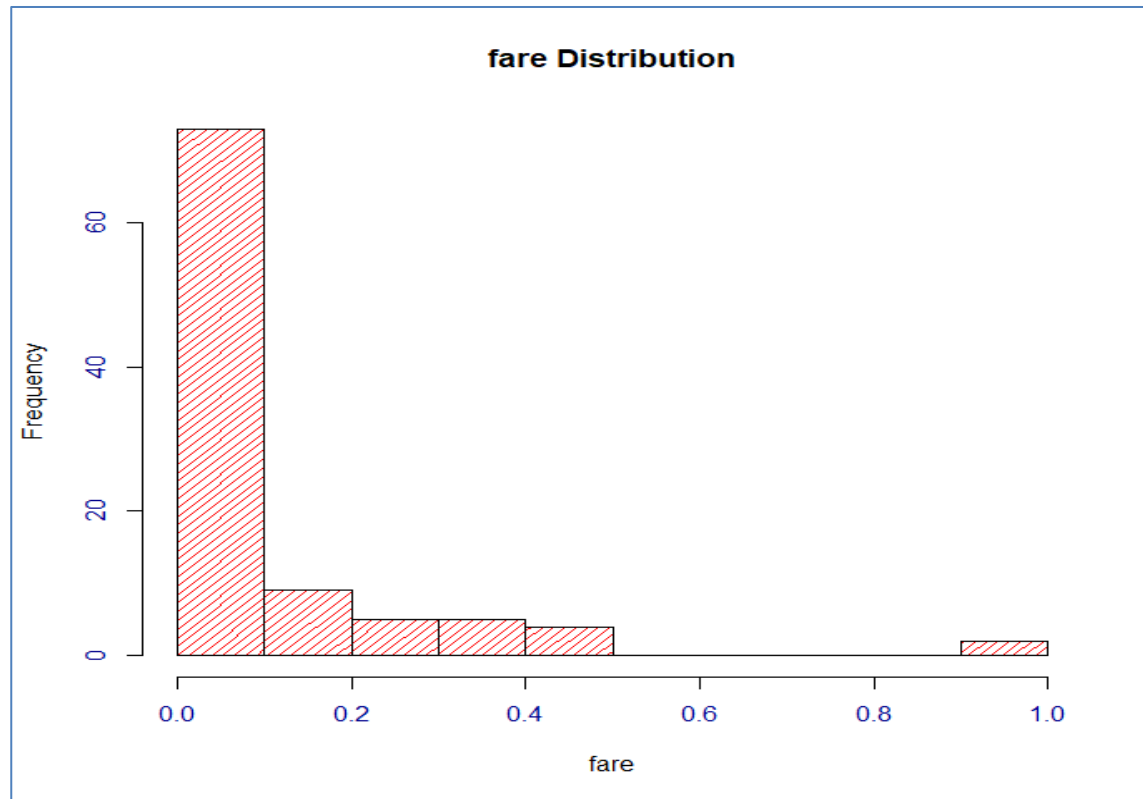
Comment: The "survived" column was checked for missing values, and none were found.

Date Visualization: **Histogram**

- For the column that has a numeric value.

```
numerical_columns <- sapply(mydata, is.numeric)
for (col in names(mydata)[numerical_columns]) {
  hist(mydata[[col]], main = paste(col, "Distribution"), xlab = col, col = "red",
    border = "black", breaks = 10, col.axis = "darkblue", density = 20)
}
```



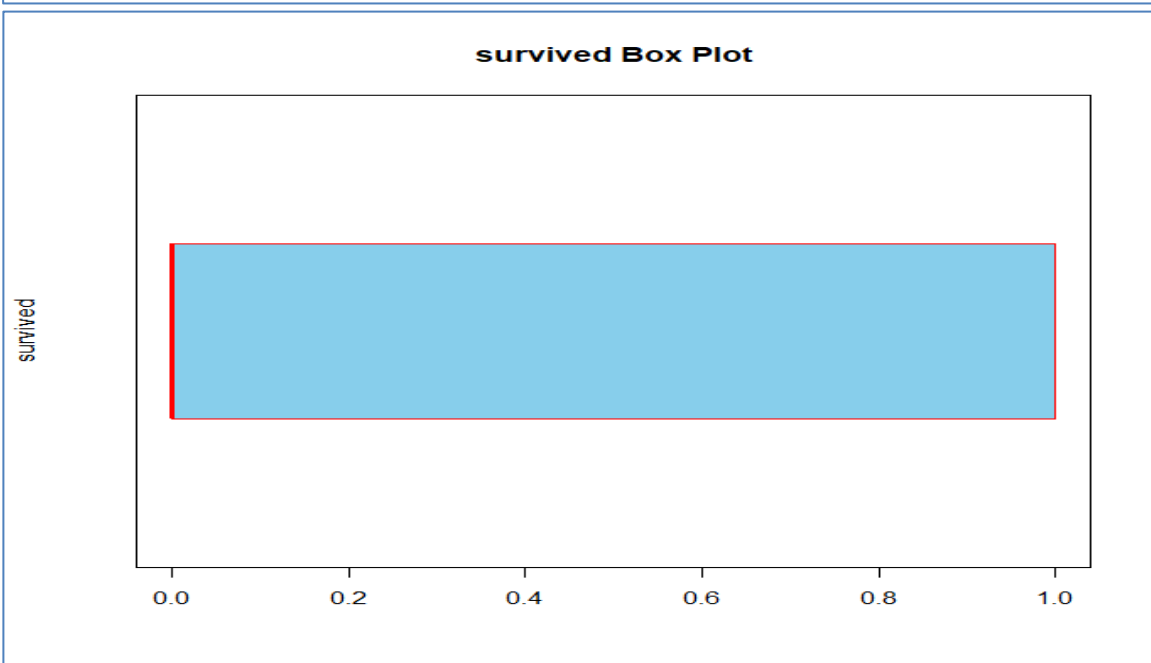
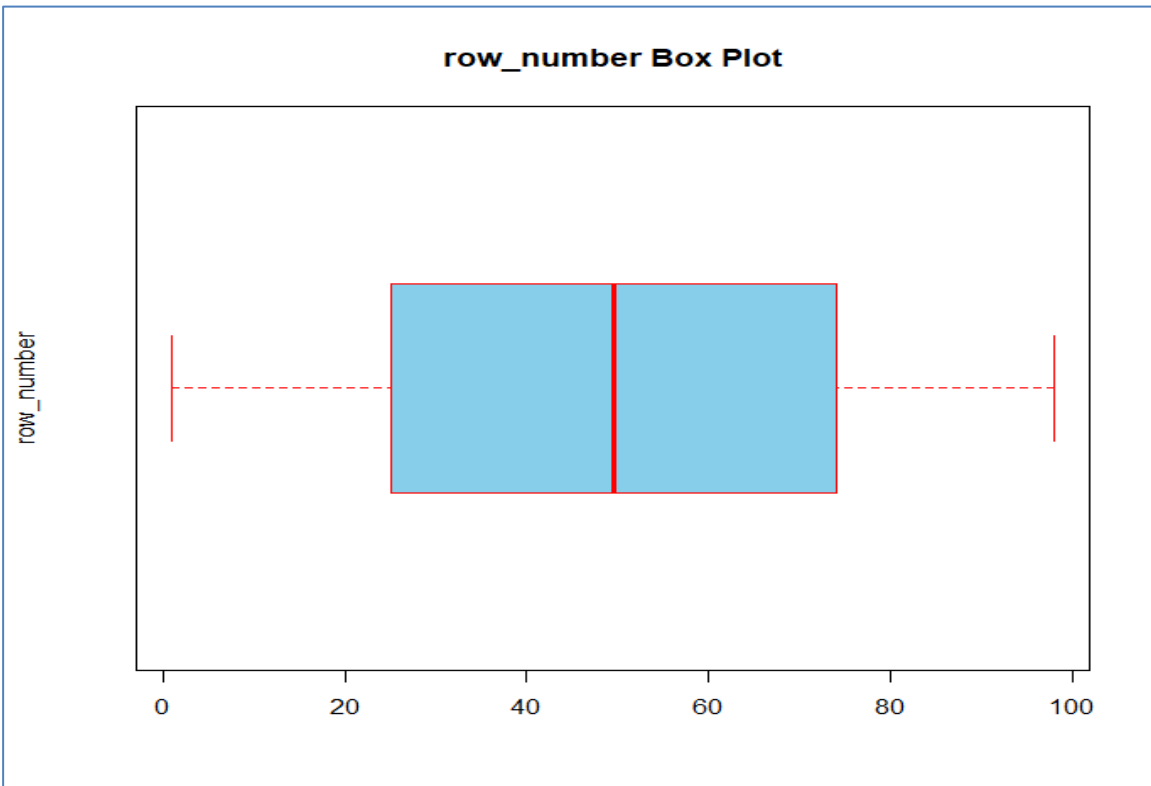


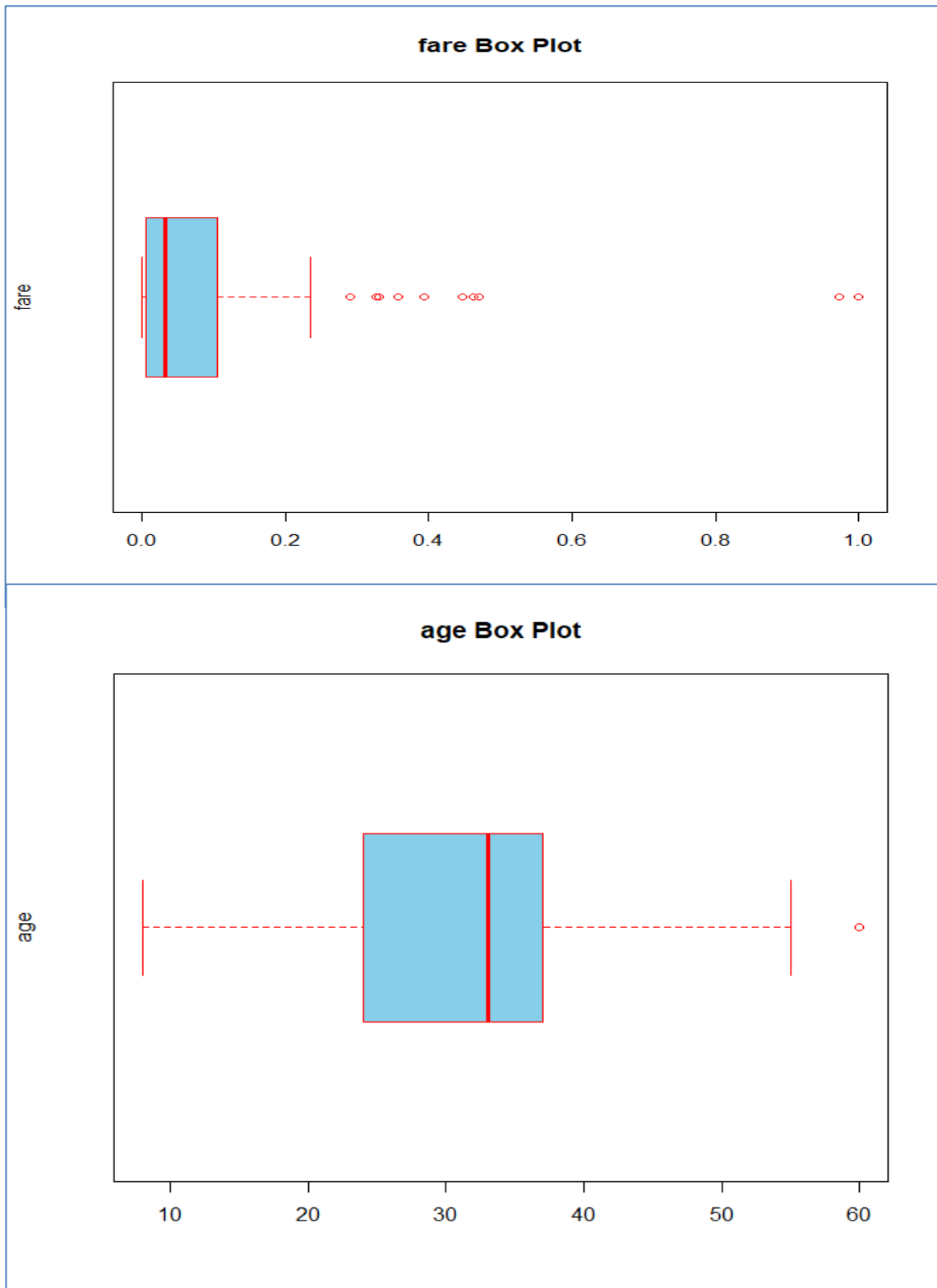
Comment: Histogram plotting was used in Numerical columns to show the data visualization through Histogram.

Date Visualization: **Boxplot**

- **For the column that has a numeric value.**

```
numerical_columns <- sapply(mydata, is.numeric)
for (col in names(mydata)[numerical_columns]) {
  boxplot(mydata[[col]], main = paste(col, "Box Plot"), ylab = col, col = "skyblue",
    border = "red", notch = FALSE, horizontal = TRUE)
}
```





Comment: Box plotting was used in Numerical columns to show the data visualization through Box plot.

Visualization: Bar-plot

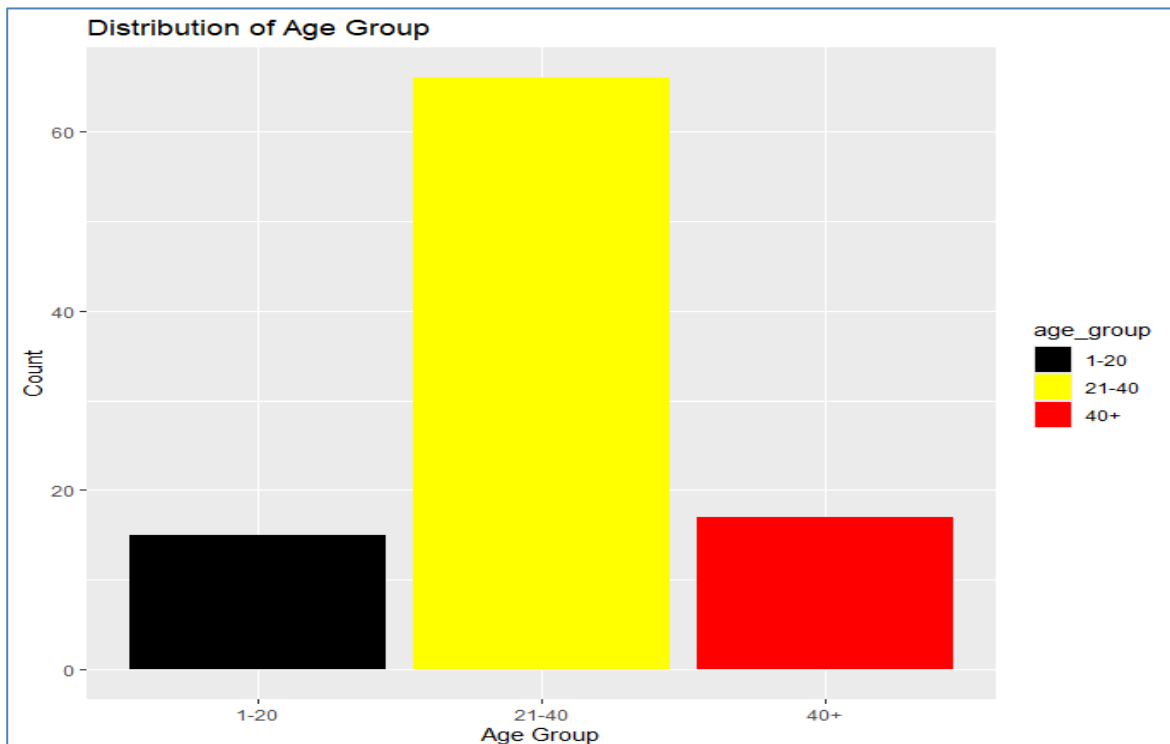
▪ Distribution of Only Age Categorical

```
mydata$age_group <- cut(mydata$age, breaks = c(0, 20, 40, Inf),
                        labels = c("1-20", "21-40", "40+"), include.lowest = TRUE)

print(head(mydata))
print(mydata)

age_plot <- ggplot(mydata, aes(x = age_group, fill = age_group)) +
  geom_bar() +
  labs(title = "Distribution of Age Group",
       x = "Age Group",
       y = "Count") +
  scale_fill_manual(values = c("1-20" = "black", "21-40" = "yellow", "40+" = "red")) +
  scale_x_discrete(labels = c("1-20" = "1-20", "21-40" = "21-40", "40+" = "40+"))

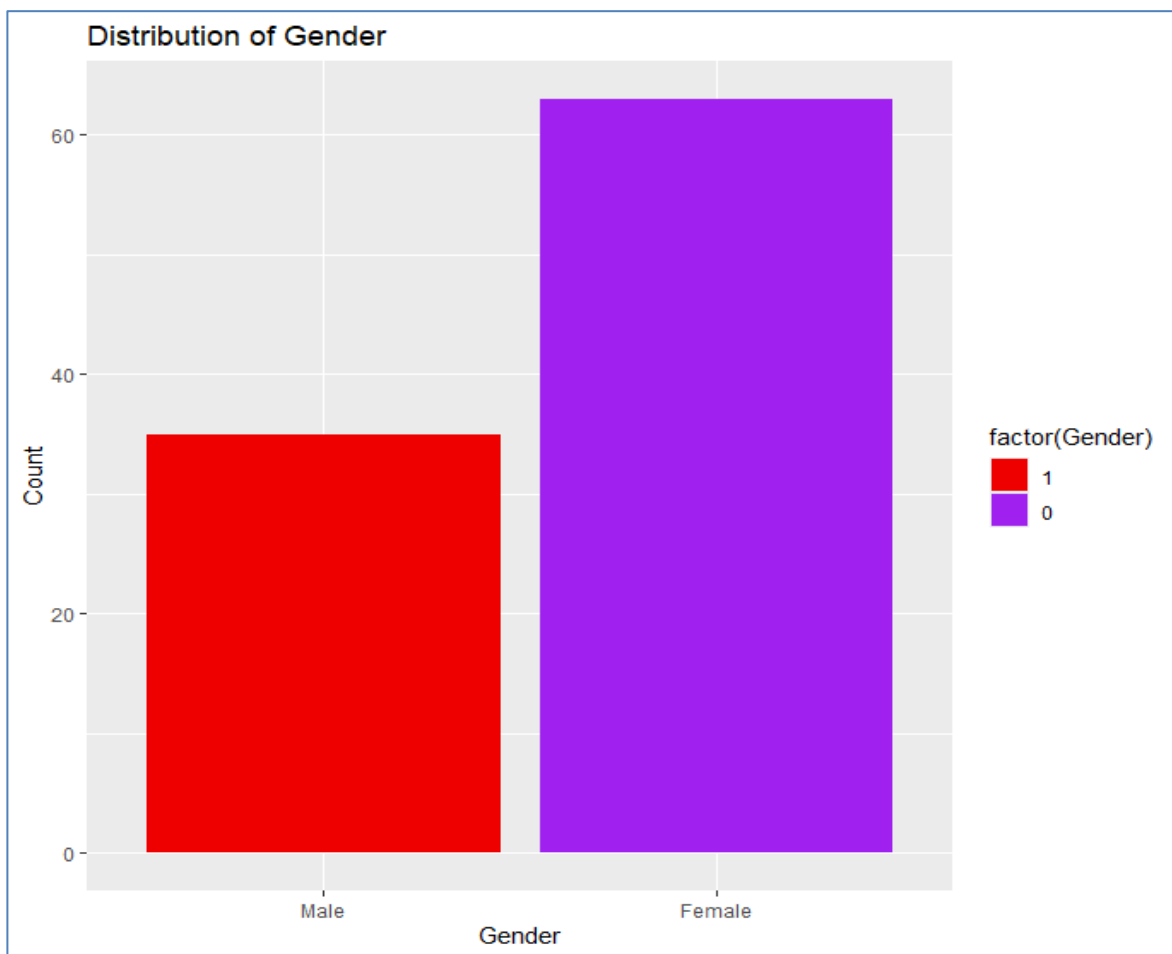
print(age_plot)
```



Comment: Age column was divided into three Age groups like 1-20, 21-40, 40+ and the data distribution is shown in the graph. People of 21-40 age can be seen most as passengers.

- **Distribution of Gender Plot**

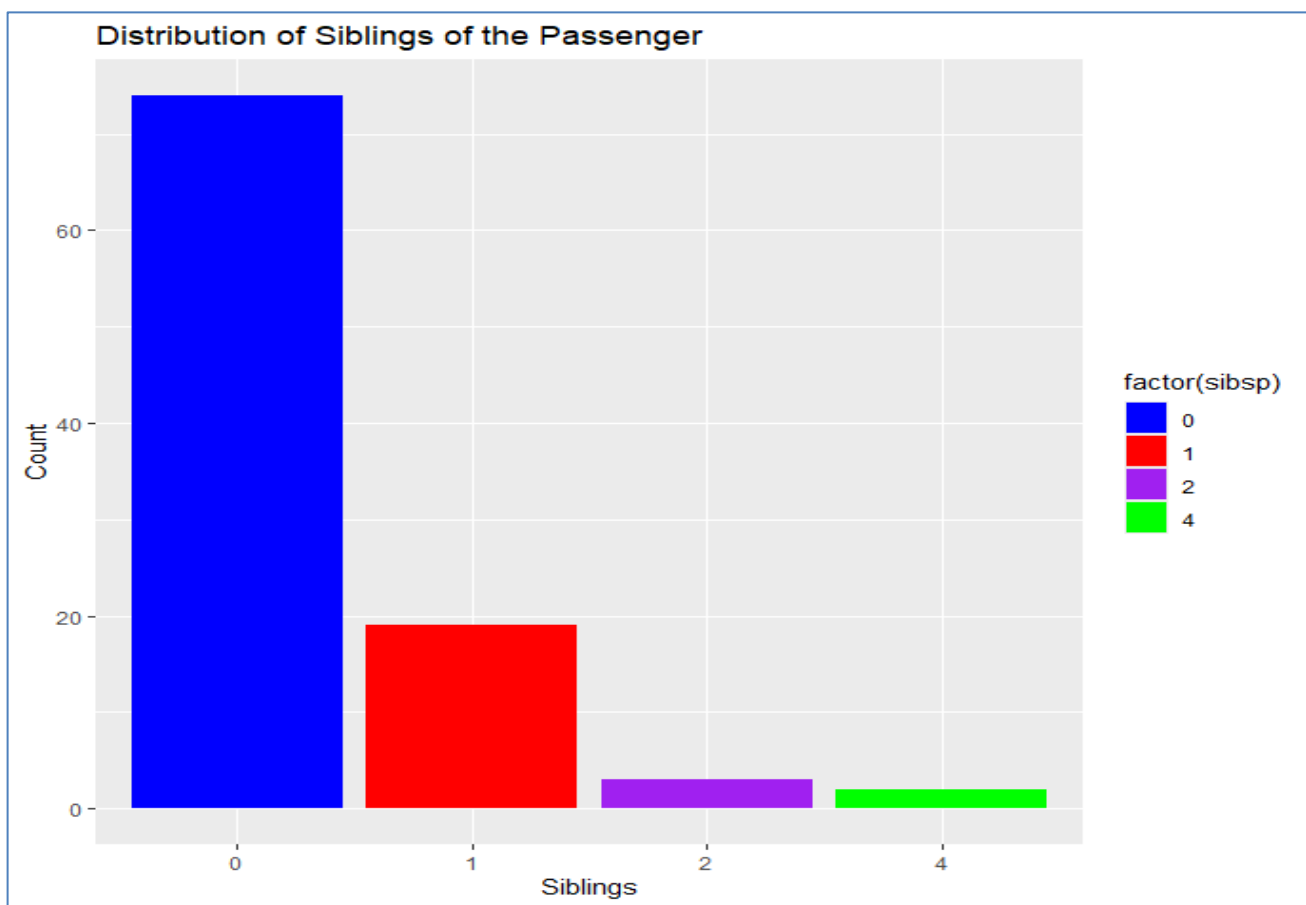
```
Gender_plot <- ggplot(mydata, aes(x = factor(Gender), fill = factor(Gender))) +  
  geom_bar() +  
  labs(title = "Distribution of Gender",  
        x = "Gender",  
        y = "Count") +  
  scale_fill_manual(values = c("1" = "red2", "0" = "purple")) +  
  scale_x_discrete(labels = c("1" = "Male", "0" = "Female"))  
print(Gender_plot)
```



Comment: Values '0' and '1' were used to show the data distribution of the Gender column in the graph with female gender being the maximum.

- **Distribution of Siblings of the Passenger**

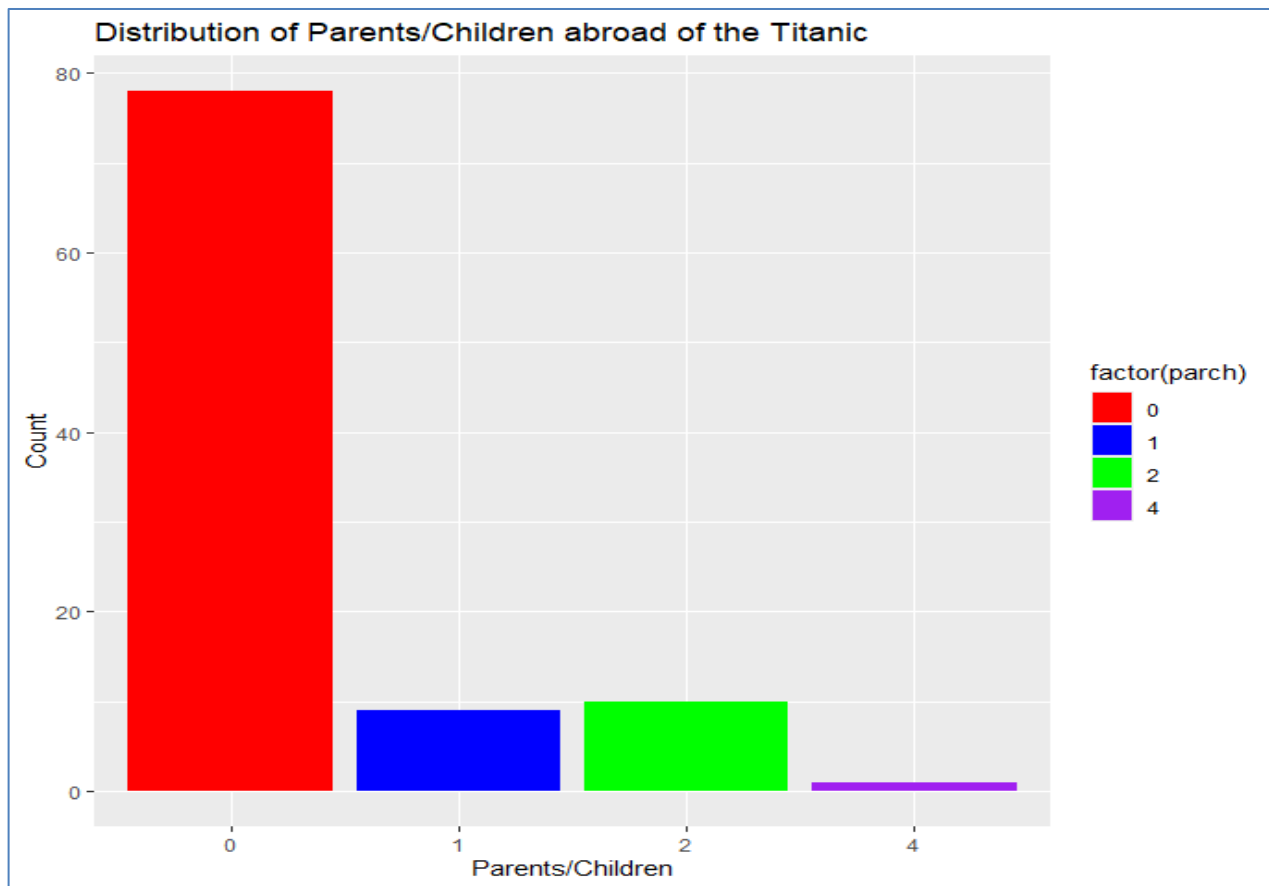
```
sibsp_plot <- ggplot(mydata, aes(x = factor(sibsp), fill = factor(sibsp))) +  
  geom_bar() +  
  labs(title = "Distribution of Siblings of the Passenger",  
        x = "Siblings",  
        y = "Count") +  
  scale_fill_manual(values = c("0" = "blue", "1" = "red", "2" = "purple", "3" = "yellow", "4" =  
"green")) +  
  scale_x_discrete(labels = c("0" = "0", "1" = "1", "2" = "2", "3" = "3", "4" = "4" ))  
print(sibsp_plot)
```



Comment: Values '0', '1', '2', '3' and '4' were used to show the data distribution of the siblings of the passenger (sibsp) with 0 being the maximum and 4 being the minimum.

- **Distribution of Parents/Children abroad of the Titanic**

```
parch_plot <- ggplot(mydata, aes(x = factor(parch), fill = factor(parch))) +  
  geom_bar() +  
  labs(title = "Distribution of Parents/Children abroad of the Titanic",  
        x = "Parents/Children",  
        y = "Count") +  
  scale_fill_manual(values = c("0" = "red", "1" = "blue", "2" = "green", "3" = "yellow", "4" =  
"purple")) +  
  scale_x_discrete(labels = c("0" = "0", "1" = "1", "2" = "2", "3" = "3", "4" = "4" ))  
print(parch_plot)
```



Comment: Values '0', '1', '2', '3' and '4' were used to show the data distribution of the Parents/Children abroad the Titanic (parch) with 0 being the maximum and 4 being the minimum.

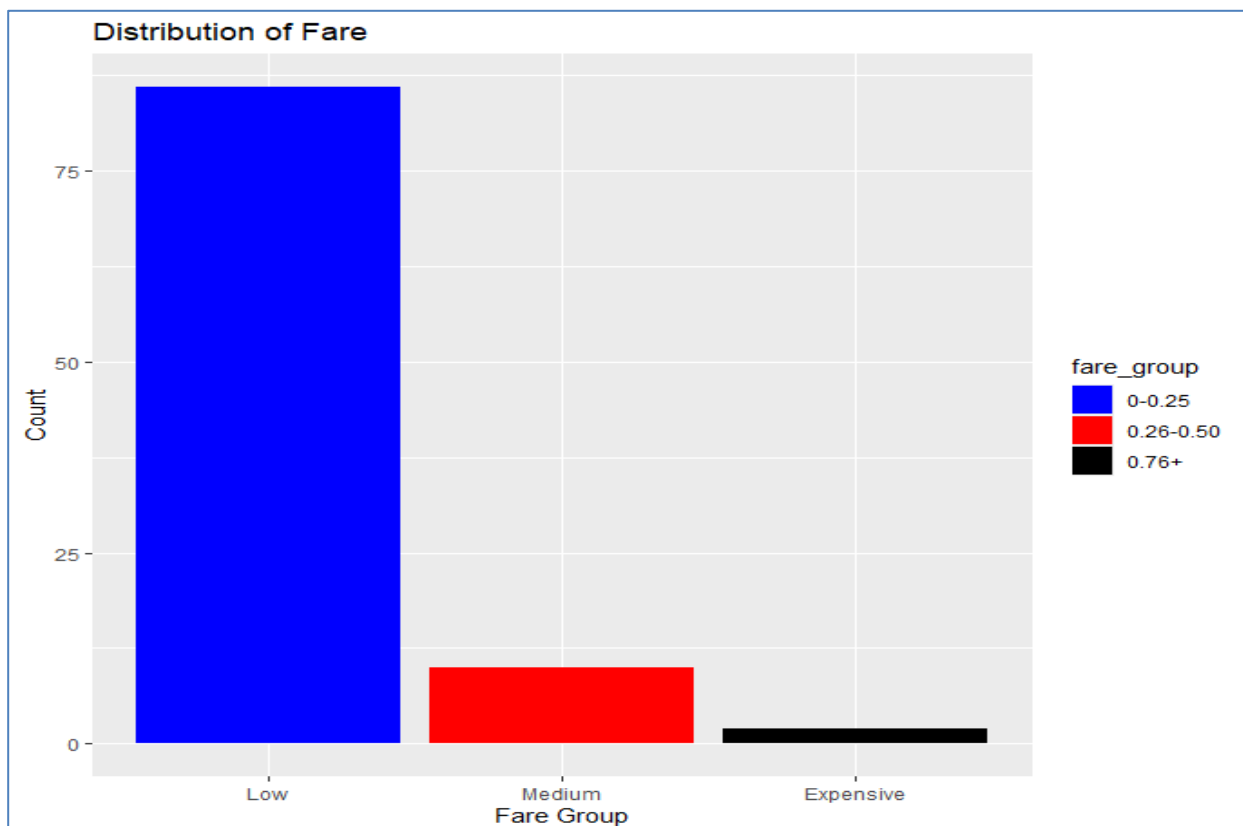
▪ Graph of Fare

```
mydata$fare_group <- cut(mydata$fare, breaks = c(0, 0.25, 0.50, 0.75, Inf),
                        labels = c("0-0.25", "0.26-0.50", "0.51-0.75", "0.76+"), include.lowest =
TRUE)

print(head(mydata))
print(mydata)

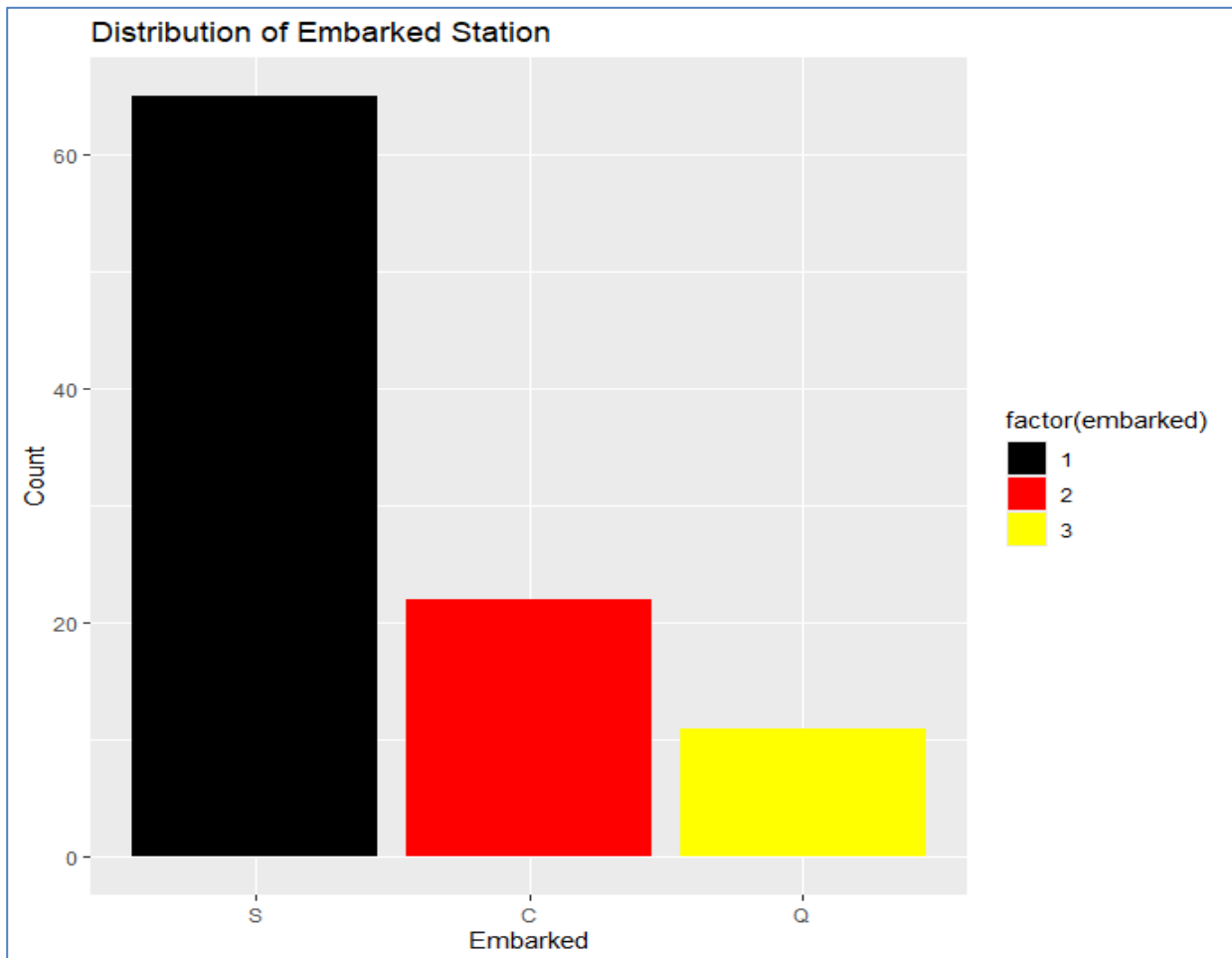
fare_plot <- ggplot(mydata, aes(x = fare_group, fill = fare_group)) +
  geom_bar() +
  labs(title = "Distribution of Fare",
       x = "Fare Group",
       y = "Count") +
  scale_fill_manual(values = c("0-0.25" = "blue", "0.26-0.50" = "red", "0.51-0.75" =
"purple", "0.76+" = "black")) +
  scale_x_discrete(labels = c("0-0.25" = "Low", "0.26-0.50" = "Medium", "0.51-0.75" =
"High", "0.76+" = "Expensive"))

print(fare_plot)
```



▪ Embarked Plot

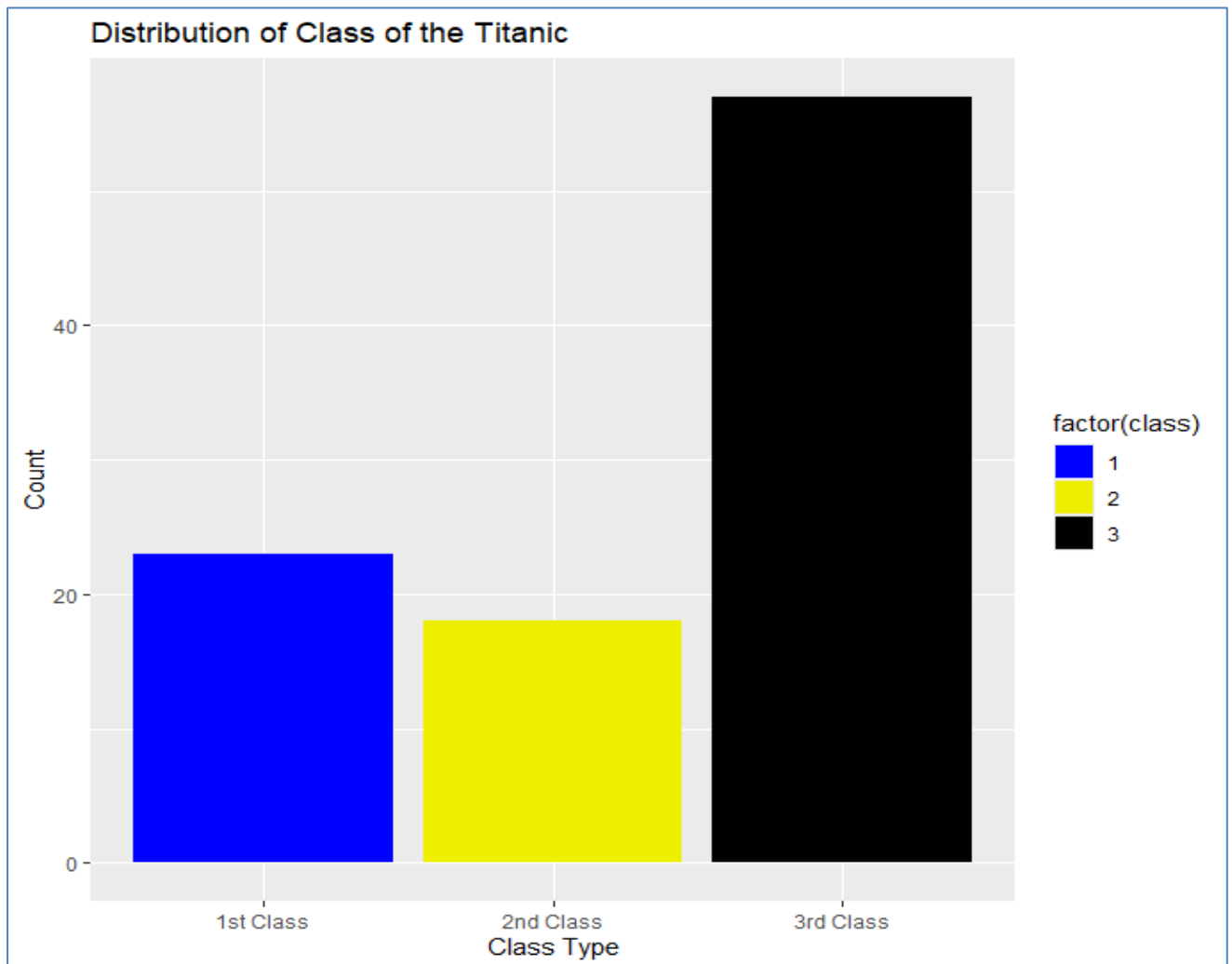
```
embarked_plot <- ggplot(mydata, aes(x = factor(embarked), fill = factor(embarked))) +  
  geom_bar() +  
  labs(title = "Distribution of Embarked Station",  
        x = "Embarked",  
        y = "Count") +  
  scale_fill_manual(values = c("1" = "black", "2" = "red", "3" = "yellow")) +  
  scale_x_discrete(labels = c("1" = "S", "2" = "C", "3" = "Q"))  
print(embarked_plot)
```



Comment: Values '1' = S, '2' = C, '3' = Q were used to show the data distribution of the Embarked Station of Titanic Passengers with S being the maximum and Q being the minimum.

- **Class Plot**

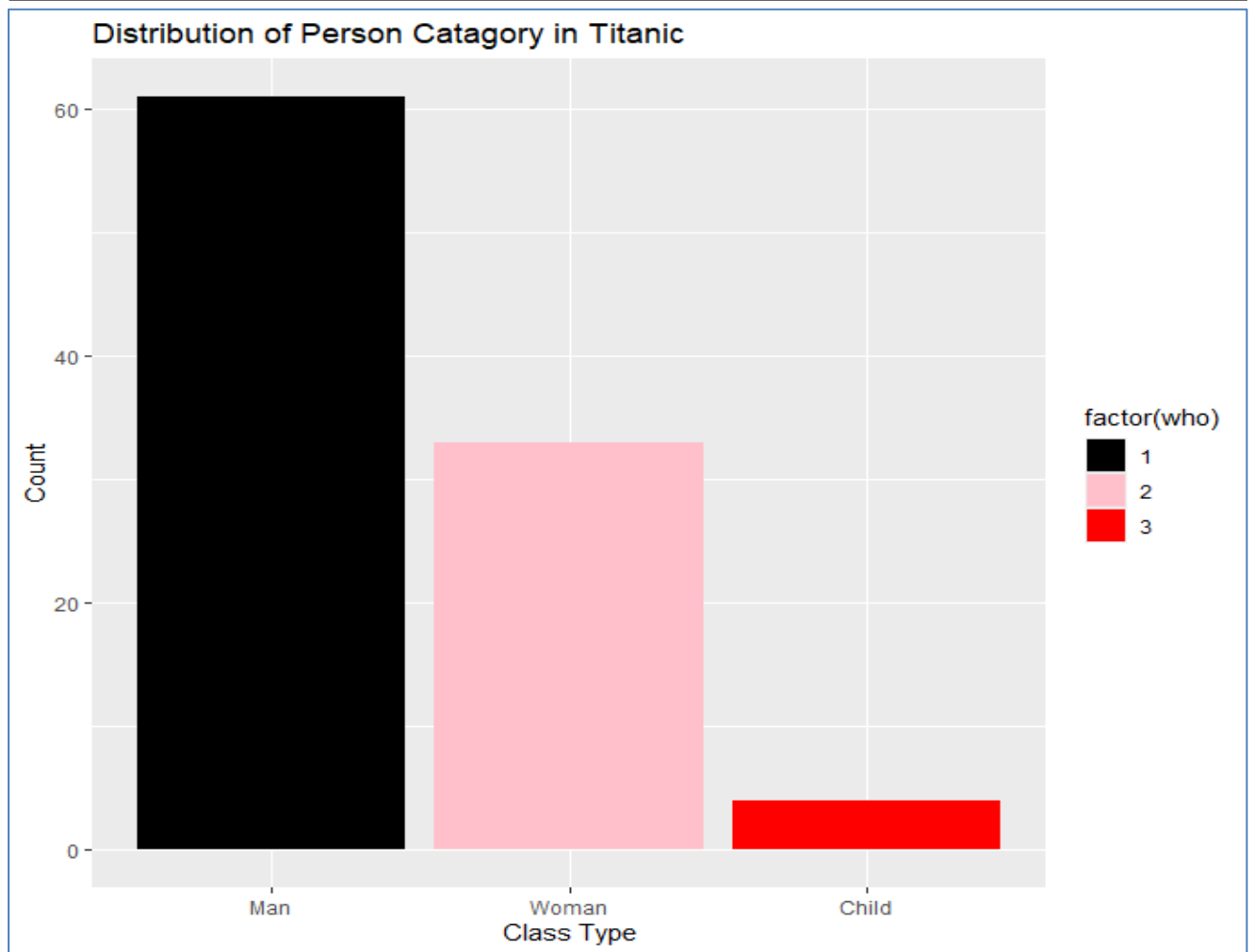
```
class_plot <- ggplot(mydata, aes(x = factor(class), fill = factor(class))) +  
  geom_bar() +  
  labs(title = "Distribution of Class of the Titanic",  
        x = "Class Type",  
        y = "Count") +  
  scale_fill_manual(values = c("1" = "blue", "2" = "yellow2", "3" = "black")) +  
  scale_x_discrete(labels = c("1" = "1st Class", "2" = "2nd Class", "3" = "3rd Class"))  
print(class_plot)
```



Comment: Values '1', '2' and '3' were used to show the data distribution of the Class of the Titanic with 3rd Class passengers being the maximum and 1st Class passengers being the minimum.

- **Who Plot**

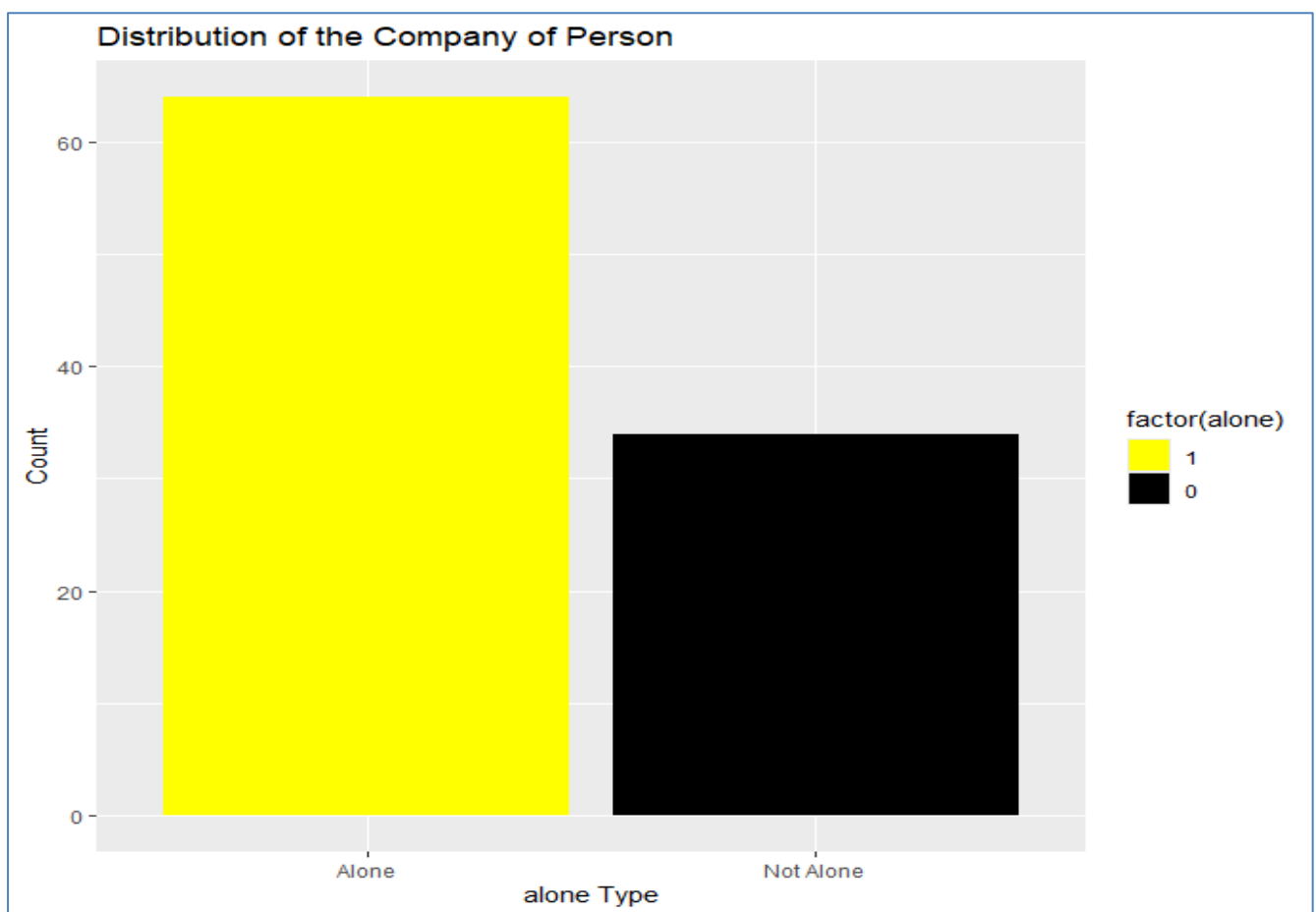
```
who_plot <- ggplot(mydata, aes(x = factor(who), fill = factor(who))) +  
  geom_bar() +  
  labs(title = "Distribution of Person Catagory in Titanic",  
        x = "Class Type",  
        y = "Count") +  
  scale_fill_manual(values = c("1" = "black", "2" = "pink", "3" = "red")) +  
  scale_x_discrete(labels = c("1" = "Man", "2" = "Woman", "3" = "Child"))  
print(who_plot)
```



Comment: Values '1', '2' and '3' were used to show the data distribution of the Person of the Titanic with Man being the maximum and Child being the minimum.

- **Alone Plot**

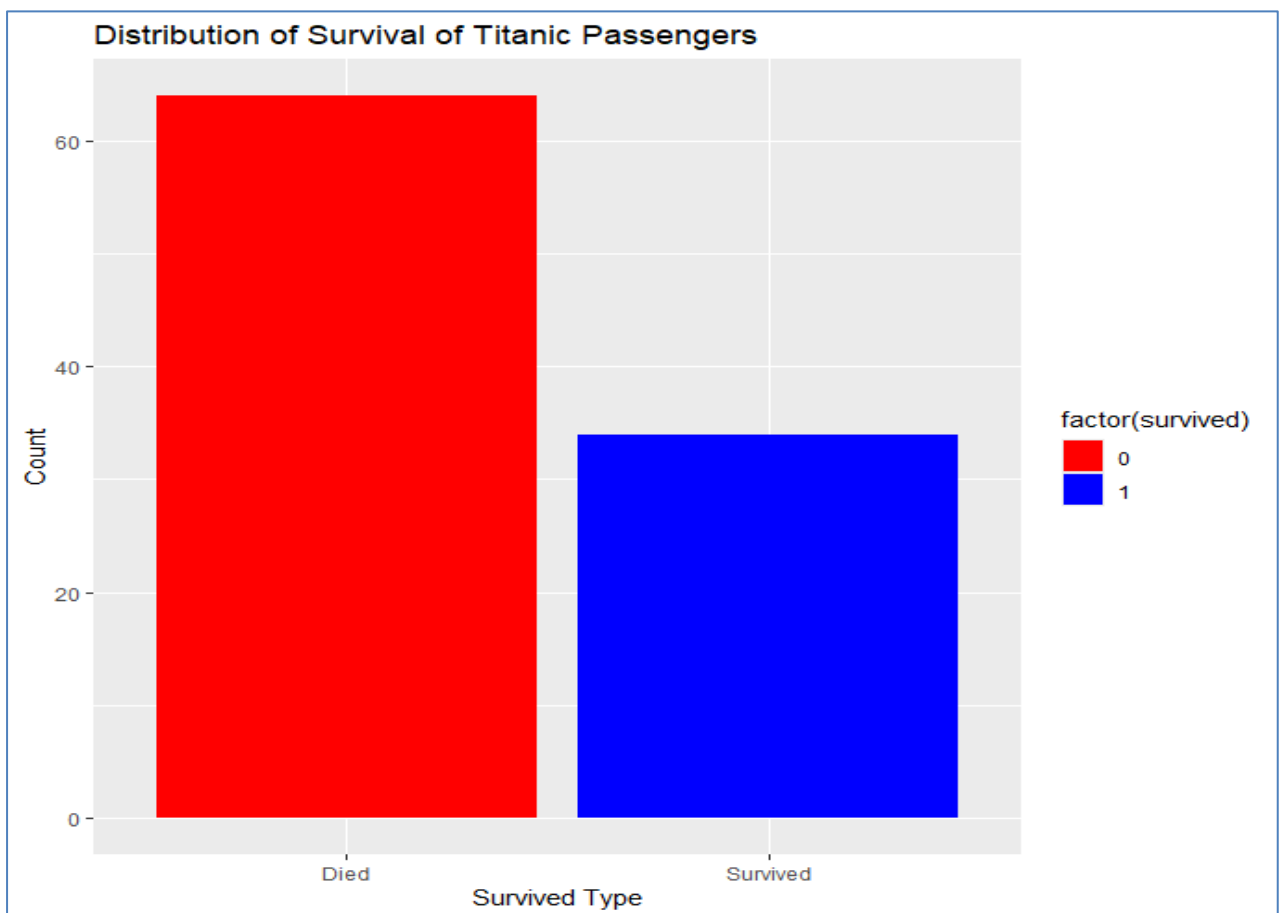
```
alone_plot <- ggplot(mydata, aes(x = factor(alone), fill = factor(alone))) +  
  geom_bar() +  
  labs(title = "Distribution of the Company of Person",  
        x = "alone Type",  
        y = "Count") +  
  scale_fill_manual(values = c("1" = "yellow", "0" = "black")) +  
  scale_x_discrete(labels = c("1" = "Alone", "0" = "Not Alone"))  
print(alone_plot)
```



Comment: The graph demonstrates if a passenger is Alone or Not Alone as Alone = '1' and Not Alone = '0'. Alone passengers were the maximum as shown in graph.

- **Survived Plot**

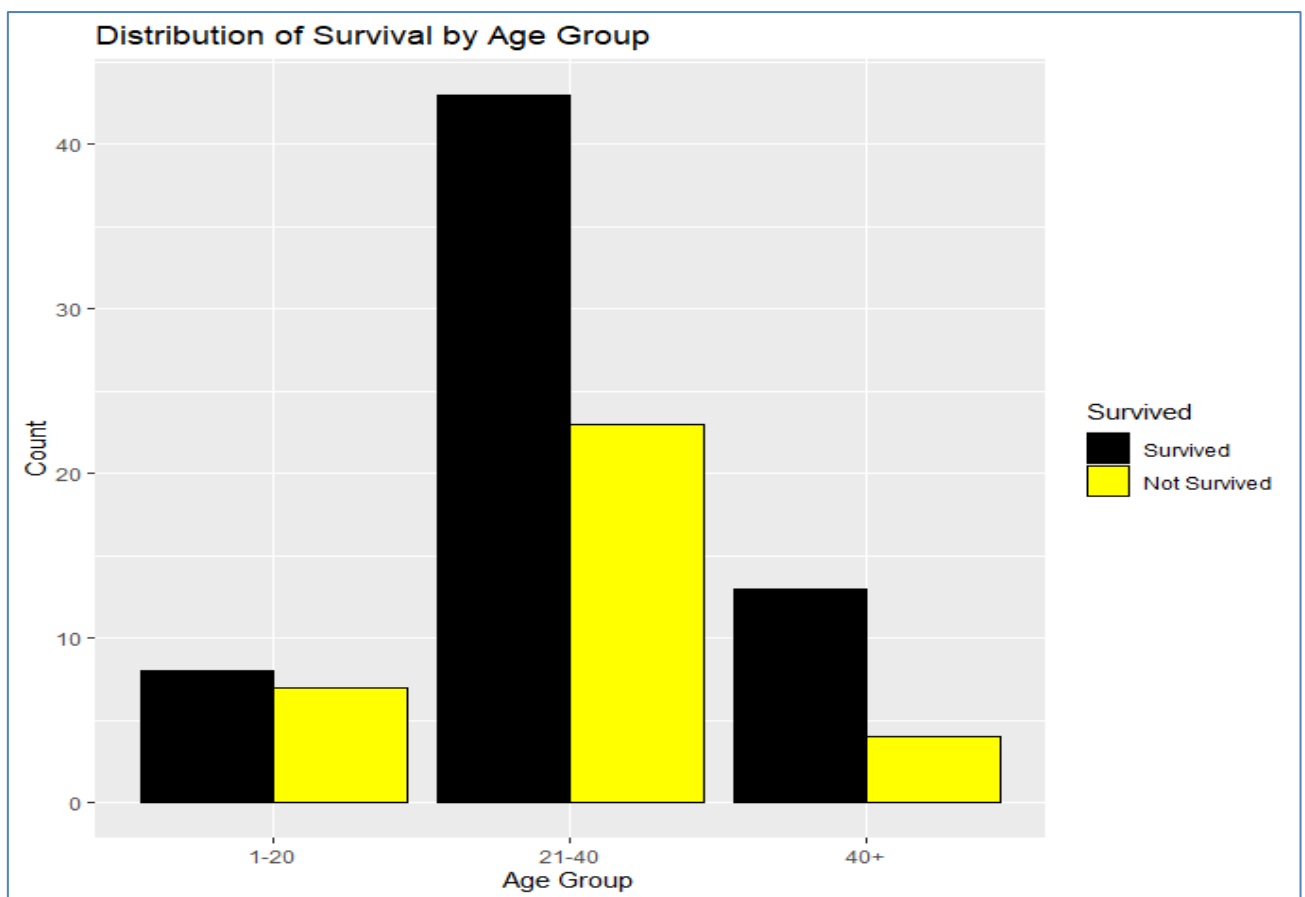
```
survived_plot <- ggplot(mydata, aes(x = factor(survived), fill = factor(survived))) +  
  geom_bar() +  
  labs(title = "Distribution of Survival of Titanic Passengers",  
        x = "Survived Type",  
        y = "Count") +  
  scale_fill_manual(values = c("1" = "blue", "0" = "red")) +  
  scale_x_discrete(labels = c("1" = "Survived", "0" = "Died"))  
print(survived_plot)
```



Comment: The graph demonstrates the survival of the Titanic passengers as Dies = '0' and Survived = '1'. Most of the passengers dies as shown in the graph.

- **Relation of Survival with Age**

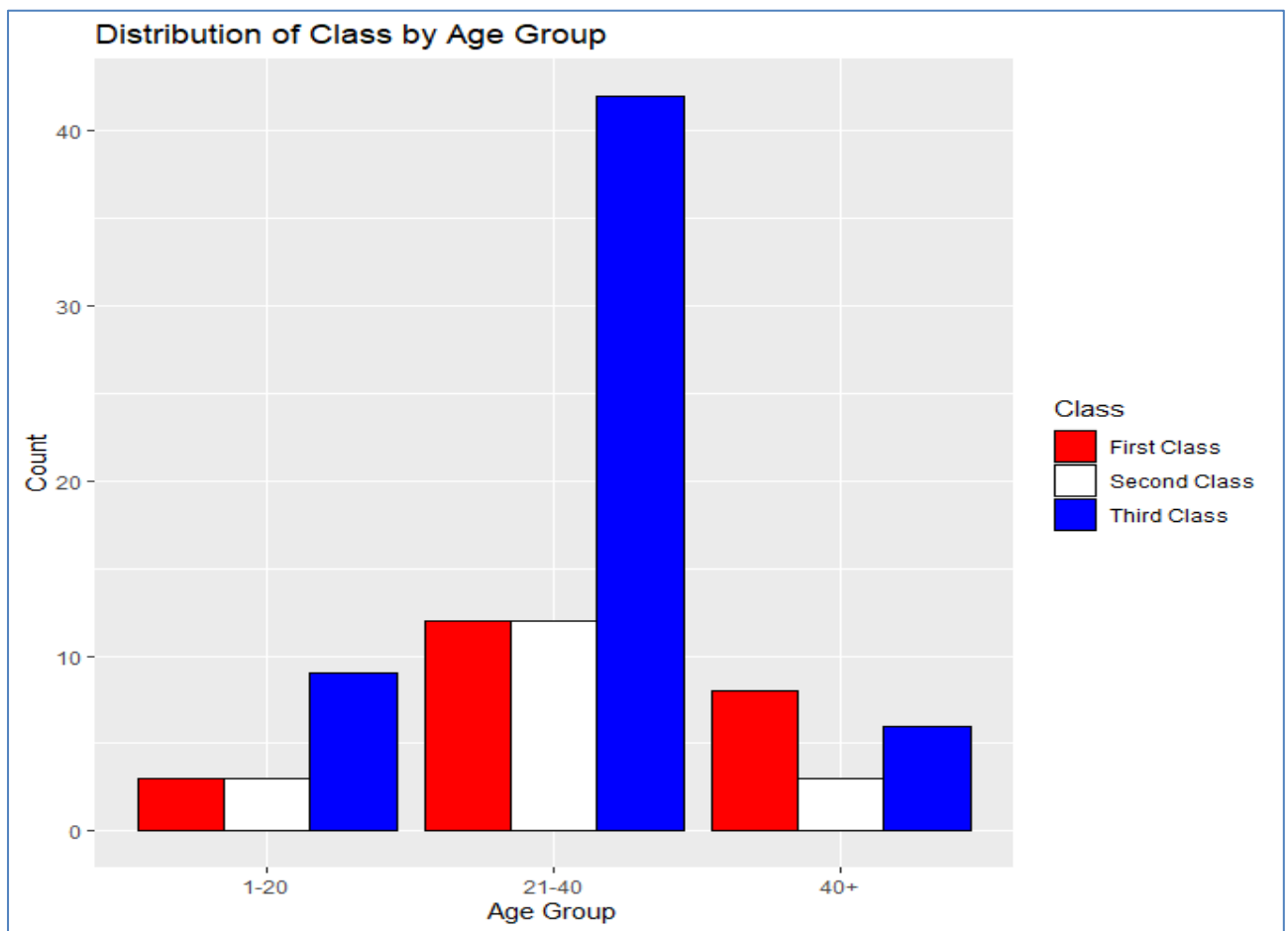
```
age_plot <- ggplot(mydata, aes(x = age_group, fill = factor(survived))) +  
  geom_bar(position = "dodge", color = "black") +  
  labs(title = "Distribution of Survival by Age Group",  
       x = "Age Group",  
       y = "Count",  
       fill = "Survived") +  
  scale_fill_manual(values = c("1" = "yellow", "0" = "black"), labels = c("Survived", "Not  
Survived"))  
  
print(age_plot)
```



Comment: The graph is a correlation of two columns 'age' and 'survived'. The graph shows the survival rate between different age groups and survival rate is most among the middle age group.

- **Relation of Class with Age**

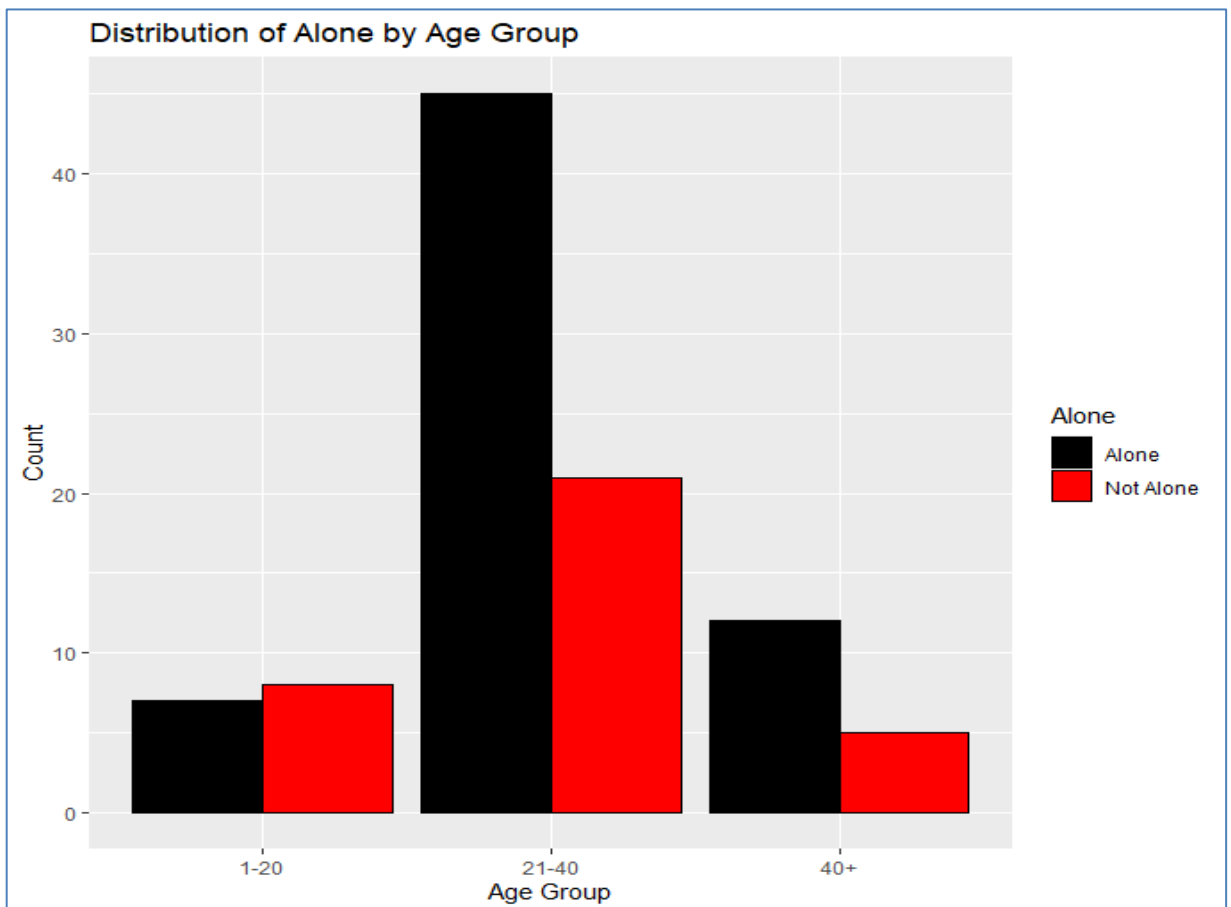
```
age_plot <- ggplot(mydata, aes(x = age_group, fill = factor(class))) +  
  geom_bar(position = "dodge", color = "black") +  
  labs(title = "Distribution of Class by Age Group",  
        x = "Age Group",  
        y = "Count",  
        fill = "Class") +  
  scale_fill_manual(values = c("1" = "red", "2" = "white", "3" = "blue"), labels = c("First Class",  
"Second Class", "Third Class"))
```



Comment: The graph is a correlation of two columns 'age' and 'class'. The graph shows the ticket of different classes bought by various age groups and third-class ticket is bought the maximum in all age groups.

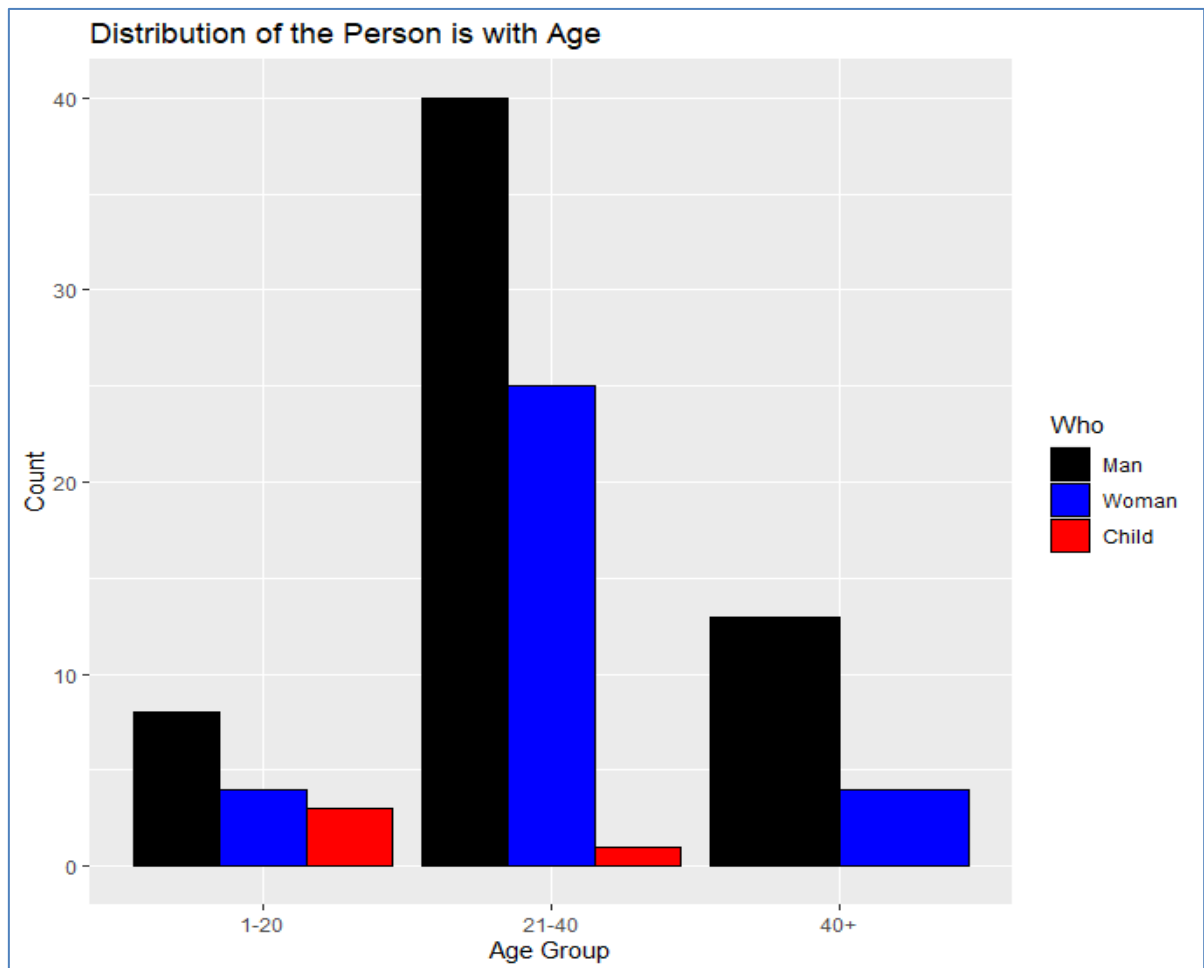
▪ Relation of Alone with Age

```
age_plot <- ggplot(mydata, aes(x = age_group, fill = factor(alone))) +  
  geom_bar(position = "dodge", color = "black") +  
  labs(title = "Distribution of Alone by Age Group",  
        x = "Age Group",  
        y = "Count",  
        fill = "Alone") +  
  scale_fill_manual(values = c("0" = "red", "1" = "black"), labels = c("Alone", "Not  
Alone"))  
  
print(age_plot)
```



▪ Relation of Age with Who

```
age_plot <- ggplot(mydata, aes(x = age_group, fill = factor(who))) +  
  geom_bar(position = "dodge", color = "black") +  
  labs(title = "Distribution of the Person is with Age",  
        x = "Age Group",  
        y = "Count",  
        fill = "Who") +  
  scale_fill_manual(values = c("1" = "black", "2" = "blue", "3" = "red"), labels = c("Man",  
      "Woman", "Child"))  
print(age_plot)
```



Comment: The graph is a correlation of two columns 'age' and 'person'. It shows the age group of people and in an age group which type of person are the most. Like, in low age group, men are the most while children are the least.

Descriptive Statistics: **Standard deviation & Mean-Median Visualize**

- **Standard Deviation**

```
numerical_columns <- sapply(mydata, is.numeric)
standard_deviations <- numeric(length = sum(numerical_columns))
names(standard_deviations) <- names(mydata)[numerical_columns]

for (col in names(mydata)[numerical_columns]) {
  standard_deviations[col] <- sd(mydata[[col]], na.rm = TRUE)
}
print("Standard Deviations:")
print(standard_deviations)

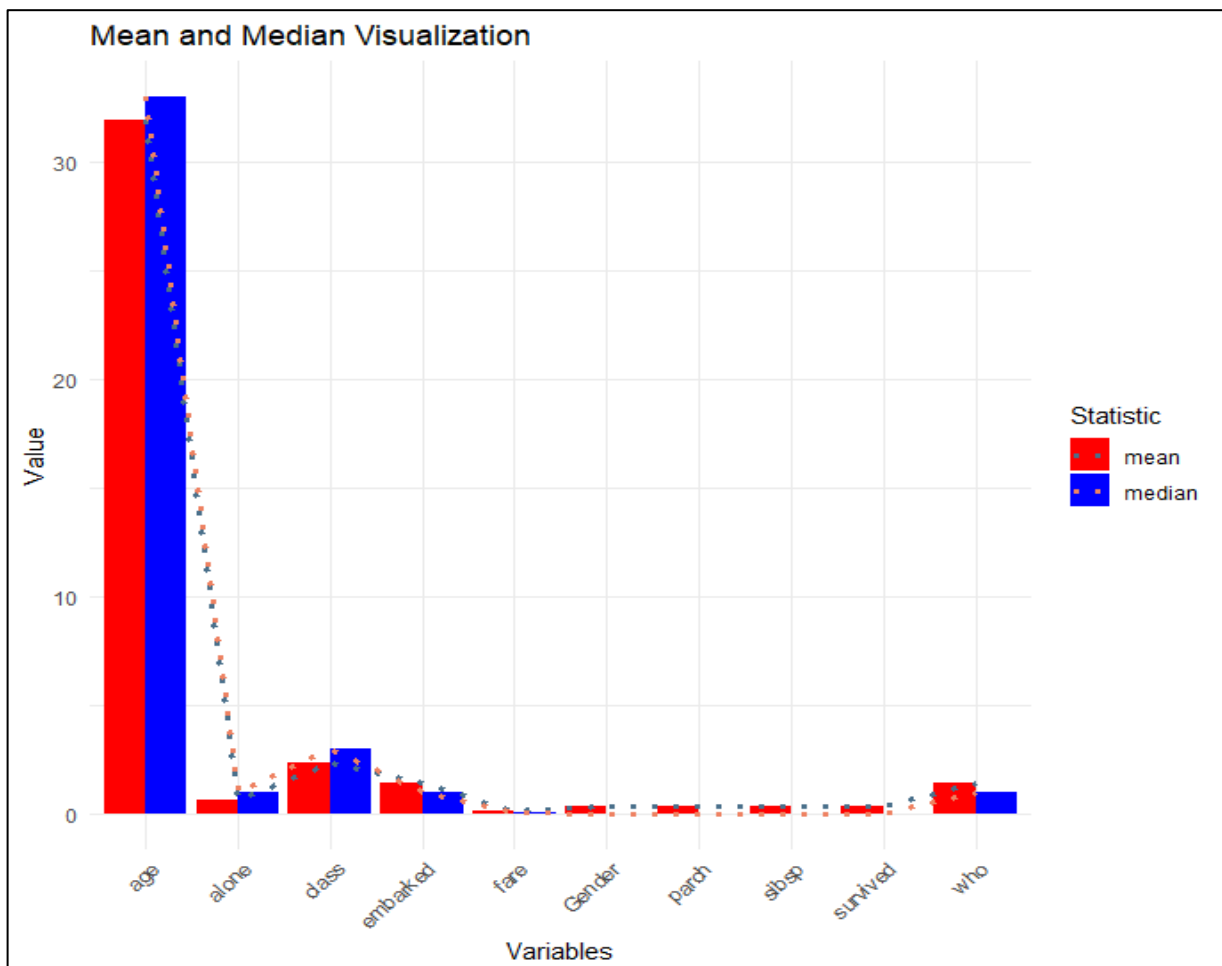
mydata <- subset(mydata, select = -row_number)
rownames(mydata) <- NULL
View(mydata)
```

```
> print("Standard Deviations:")
[1] "Standard Deviations:"
> print(standard_deviations)
      age      fare  survived row_number
10.6561753 0.1755071 0.4784434 28.4341344
>
> mydata <- subset(mydata, select = -row_number)
> rownames(mydata) <- NULL
> View(mydata)
> |
```

▪ Mean Median Plot

```
mean_numeric <- function(x) {
  if (!is.numeric(x)) {
    return(NA)
  }
  mean(x, na.rm = TRUE)
}

means <- sapply(mydata, mean_numeric)
medians <- sapply(mydata, function(x) median(x, na.rm = TRUE))
medians <- as.numeric(medians)
summary_df <- data.frame(variable = names(means),
                          mean = means,
                          median = medians)
summary_df_long <- pivot_longer(summary_df, cols = c(mean, median),
                                names_to = "statistic", values_to = "value")
```



Comment: The graph demonstrates the mean median visualization of each column in the Titanic dataset. As it can be seen, 'age' column differs from the rest of the column in case of mean-median value.

Dataset: **Overview Before Analysis**

Column Name [Total data 105]	Data Type	Missing Values	Invalid Value	Noisy Value	Date Type Conversion
Gender	chr	yes	no	no	yes
age	int	yes	no	yes	no
sibsp	int	yes	no	no	no
parch	int	yes	no	no	no
fare	chr	yes	yes	no	yes
embarked	chr	yes	no	no	yes
class	chr	yes	no	no	yes
who	chr	yes	yes	no	yes
alone	logi	yes	no	no	no
survived	int	no	no	no	no

Dataset: **Overview After Analysis**

Column Name [Total data 98]	Data Type	Missing Values	Invalid Value	Noisy Value	Date Type Conversion
Gender	chr	no	no	no	no
age	int	no	no	no	no
sibsp	int	no	no	no	no
parch	int	no	no	no	no
fare	chr	no	no	no	no
embarked	chr	no	no	no	no
class	chr	no	no	no	no
who	chr	no	no	no	no
alone	logi	no	no	no	no
survived	int	no	no	no	no